

DIRAC

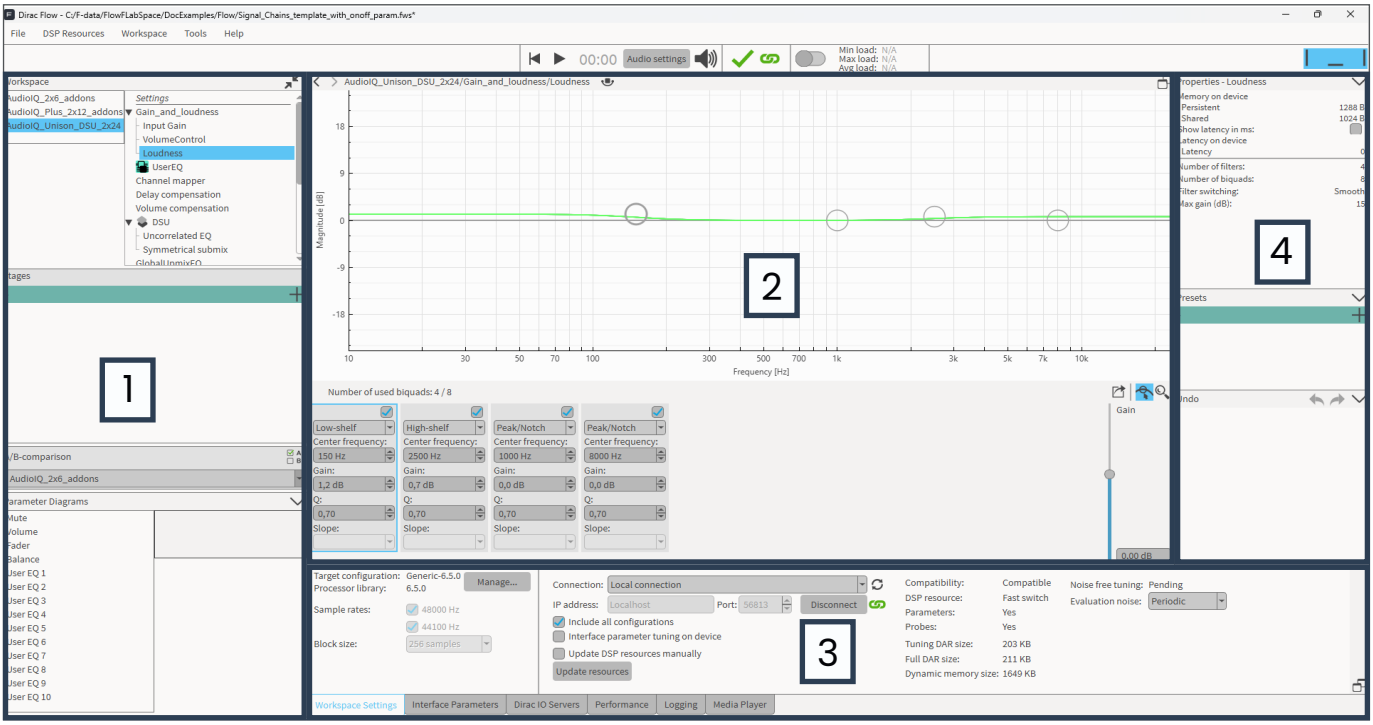
Flow 5.6

DPL 6.6

User manual

Contents

1. Introduction 3
1.1. About Dirac Flow 3
1.2. System requirements 3
1.3. License 3
1.4. Names of modules and processors 3
1.5. Interface 3
1.6. Your Workspace 4
1.7. Theme 5
1.8. Software update for Flow 5
2. Power EQ 6
2.1. Overview 6
2.2. To set biquad filter values 6
2.3. Tools 6
3. Filter inspection 7
3.1. Overview 7
3.2. To import a filter 7
3.3. To inspect a filter 7
3.4. To export a filter 7
3.5. To remove a filter 7
4. Channel EQ 8
4.1. Overview 8
4.2. To create/import a filter 8
4.3. To edit a filter 8
4.4. To duplicate a filter 9
4.5. To export a filter 9
4.6. To remove a filter 9
5. Matrix mixer 10
5.1. Overview 10
5.2. To create/import a filter 10
5.3. To edit a filter 10
5.4. To duplicate a filter 11
5.5. To export a filter 11
5.6. To remove a filter 11
6. Dirac IO 12
6.1. Overview 12
6.2. To import a filter with Dirac IO 12
6.3. To edit a filter 13
7. Interface parameters 14
7.1. Overview 14
7.2. Parameters 14
7.3. To adjust an interface parameter 14
7.4. To adjust processor settings 14
8. Device connection 17
8.1. Overview 17
8.2. To connect 17
8.3. Connection parameters 18
8.4. Performance measurement 18
8.5. Logging 19
8.6. To disconnect 19
9. DSP Resources export 20
9.1. Overview 20
9.2. To export DSP resources 20
9.3. To do a quick export 20
9.4. To do an export with specific settings 20
10. Audio settings 22
10.1. Overview 22
10.2. Audio settings panels 22
11. Utility modules 24
11.1. Overview 24
11.2. Volume 24
11.3. Delay 24
11.4. Channel Mapper 24
11.5. Clipper 24
11.6. Delay compensation 24
11.7. Custom Panel 25
11.8. A/B Comparison 25
11.9. Media Player 25
12. Compare workspaces 26
12.1. Overview 26
12.2. Diff tool 26
12.3. To compare two workspaces 26
12.4. To compare two configurations 26
13. Symmetrical submixer 27
13.1. Overview 27
13.2. To create new/import a filter 27
13.3. Spatial settings 27
13.4. To edit a filter 27
13.5. To duplicate a filter 28
13.6. To export a filter 28
13.7. To remove a filter 28
14. Peak limiter 29
14.1. Overview 29
14.2. Setup 29
14.3. Settings 29
14.4. Level meters 29
15. Reverb 30
15.1. Overview 30
15.2. Settings 30
15.3. Level meters 30
15.4. Properties 30
16. AGC 31
16.1. Overview 31
16.2. Setup 31
16.3. Settings 31
16.4. Level meters 31



1. Workspace panel
2. Processor panel
3. Settings panel
4. Properties panel

1. Introduction

1.1. About Dirac Flow

Dirac Flow is an intuitive tool for real-time tuning of specific Dirac features. It controls the runtime on the target system with minimal delay, so you can hear the effect of each change immediately. This enables an efficient workflow where you can quickly adjust and refine the sound. Which features are available depends on what has been licensed and enabled for your project in your workspace. Some features are configured by Dirac only, while others are exposed for you to adjust. This document explains how to use Dirac Flow as delivered in your setup.

1.2. System requirements

Dirac Flow requires a Windows OS computer with Microsoft Windows 11 or later as operating system.

If you have an older version of Windows OS, speak to Dirac to check compatibility.

1.3. License

To access Flow and its features, you must have a software license issued for each seat (named individual). Each license is valid for a specified individual over a specified time period.

1.4. Names of modules and processors

The names of the workspaces, modules, processors and parameters in your workspace may not have the exact same name as the modules and processors described and displayed in this manual.

In many cases, this documentation gives instructions for a type of processor. These instructions apply to a processor of that type that has been given a name that is applied to your specific workspace and area of use. The names and application areas of modules and processors are made according to the specification of your individual project.

1.5. Interface

The Flow interface has 4 panels:

1. Workspace panel
2. Processor panel
3. Settings panel
4. Properties panel

The workspace, settings and properties panels can be hidden from view with the toggle icons in the upper right corner.





A panel that is shown is displayed as blue on the icon, and a panel that is hidden is displayed as grey on the icon.

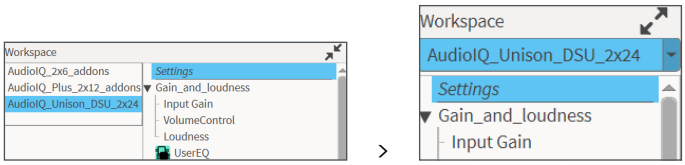
Press the left side of the icon to hide/unhide the workspace panel.

Press the middle part of the icon to hide/unhide the Settings panel.

Press the right side of the icon to hide/unhide the properties panel.

You can resize your panels by dragging the panel border toward the processor panel.

The workspace panel can be compressed to a single column panel by pressing the Compress icon and it can be expanded back to a two-column view by pressing the Expand icon .



1.5.1. Workspace panel

The workspace panel (1) has all the configurations, settings, processors, stages, A/B comparison tool, and parameter diagrams.

It is in the workspace panel that you select the configuration you want to use, select which processor you want to inspect/edit, and select which parameter diagrams you want to inspect to verify the setup.

1.5.2. Processor panel

The processor panel (2) is where the processors, settings, diagrams etc. are shown. It is the principal working area that is always displayed.

1.5.3. Settings panel

In the settings panel (3) a number of settings, interface parameters, and connections are handled. This is also the area where the performance can be measured, and logging of events in the connections can be done in case of connection issues. This is also the location of the Media Player, the built-in audio file player that can be used for playback when tuning.

Each tab holds a separate setup of parameters. Refer to the chapters in this manual for more information.

1.5.4. Properties panel

The properties panel (4) shows the properties of the processor, and if applicable, it shows presets, the filters of the processor including the filter that is sent to the device, filter information, and any changes that have been made to a filter.

1.6. Your Workspace

To open a workspace, go to the top menu File/Open and select the .fws file you want to work with.

The workspace contains the specified number of configurations and show the specified processors of the signal chain for a selected configuration.

1.6.1. To set parameter values

When a value must be set or adjusted, use one of the follwing methods to set the value. The method options can vary, and depends on the type of value that is set.

A. List value: Click on the field, choose a new value from the list and click the selected value to confirm.

B. List value: Put the mouse pointer over the field and use the mouse scroll wheel to go though the list of values one step at a time.

C. Numeric value: Click in the field and use the mouse scroll wheel up to increase the value or the scroll wheel down to decrease the value by specific increments.

D. Numeric value: Click in the field and use keyboard arrow up key to increase the value or the arrow down key to decrease the value by specific increments.

E. Numeric value: Click in the field and use the keyboard PgUp key up to increase the value or the PgDn key to decrease the value by specific increments.

F. Numeric value: If you know the exact value you want, click on the field value and type the new value. Press Enter to confirm.

G. Numeric value: Click on the field arrow up to increase the value or field arrow down to decrease the value by specific increments with each click.

1.6.2. Interface Parameters

Each of the biquad filter parameters in all processors can be made available as an interface parameter that can be changed by external input, for example a user interface.

If the settings are set as Interface parameters, they have a lock icon on them in the processor panel and their values are editable only on the Interface parameters tab in the settings panel (3).

1.6.3. Processor settings

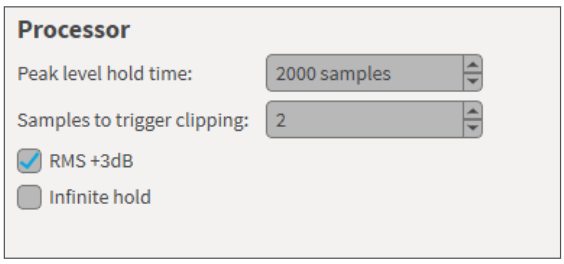
In the top menu Tools/Options you can make global settings for all the processors in Flow.

Peak level hold time sets the amount of time the peak is held after the level starts dropping.

Samples to trigger clipping sets the level for how many samples are permitted to go over the threshold before the clipping indicator shows that the threshold has been breached.

RMS +3dB (Root Mean Square) adds 3 dB to the signal energy to make it display approximately the same value as an amplitude display.

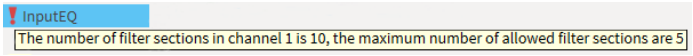
Infinite hold lets you keep the peak value infinitely and not let it fall. If you double-click on a level meter, it resets the clipping indicator and releases the hold.



1.6.4. Filter errors

If a filter has one or more errors, the processor selection in the workspace panel shows an exclamation mark.

Put the mouse pointer over the exclamation mark to show the error information. If there are several errors, the error information is for the first error. When the first error is resolved, the next error information is shown. When all errors are resolved, the exclamation mark is no longer shown, and the filter can be used for the processor.



Possible filter errors can be, for example, incorrect dimensions of the filter, unsupported sample rates, incorrect filter properties, incorrect number of layers, incorrect resampling type, incorrect number of FIR taps, or too many biquads.

1.6.5. Bypass

If a processor is to be bypassed in the signal chain, you must press the Bypass icon next to the name of the processor in the processor panel:



When the bypass is set, the bypass icon becomes orange, and the icon is also shown in front of the processor name in the signal chain:



1.6.6. Shared processors

If a processor in a workspace is shared between configurations, it displays a small icon in front of the processor name. The icon indicates the type of sharing and the channel requirements between configurations:

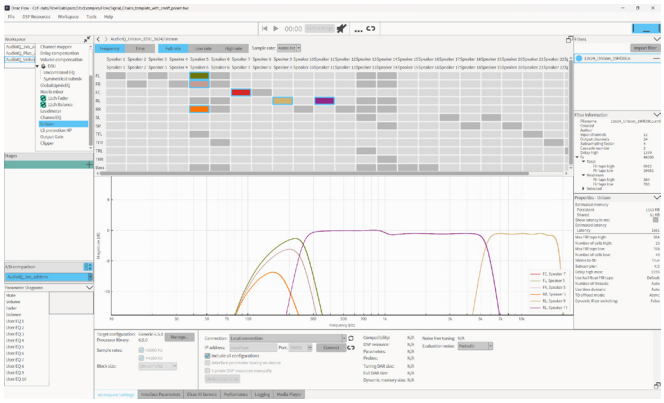
- Shared, requires the same number of channels in all configurations.
- Shared, supports different number of channels in different configurations.

1.7. Theme

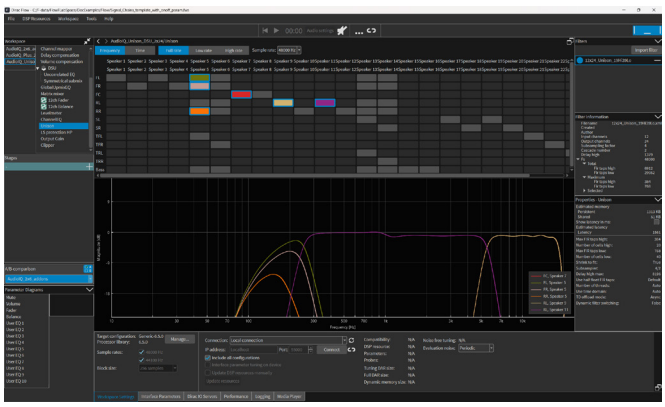
The theme of Flow can be changed from the default Light mode to Dark mode. Go to top menu Tools/Theme to select the new theme.

This user manual has screenshots and descriptions for Light mode.

If you use Flow in Dark mode, note that the colour scheme is different or inverted to what it is in Light mode. For example, if an unselected filter cell is described as “dark grey” in Light mode, the colour of the filter cell can be “light grey” in Dark mode.



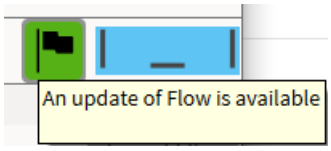
Flow in Light mode



Flow in Dark mode

1.8. Software update for Flow

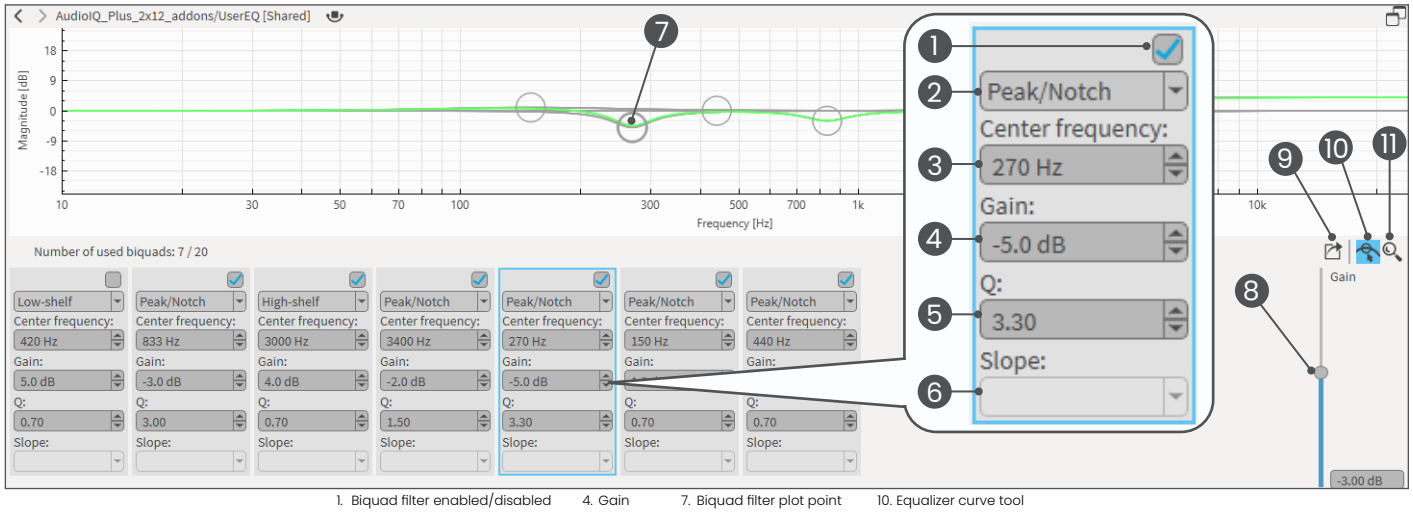
When there is a software update available for Flow, a flag appears in the top right corner of the interface.



Hover over the flag with the mouse pointer to view the update message.

Press the flag to start the update of the software.

NOTE: there is no roll-back to a previous version of Flow after an update. It’s recomended to use the latest version of Flow. It’s compatable with DPL 5.0.0 and later plus some special versions of DPL 4. Old versions of Flow may not be compatible with newer DPLs.



1. Biquad filter enabled/disabled
2. Filter type
3. Center frequency
4. Gain
5. Q-factor
6. Slope
7. Biquad filter plot point
8. Module output gain
9. Export
10. Equalizer curve tool
11. Zoom and pan tool

2. Power EQ

2.1. Overview

The **Power EQ** processor is a biquad equaliser that can be used for various purposes (User EQ, Loudness and others). The Power EQ makes it possible to set filter values for each biquad filter in the applied use of the processor in the audio system.

2.2. To set biquad filter values

Make sure that the checkbox (1) of the biquad filter is checked to enable the filter. If the checkbox is not checked, the biquad filter will not affect the output.

Click on the enabled biquad filter you want to adjust. The active biquad filter has a blue border around it and the plot point (7) in the plot area is highlighted.

You can also use the mouse pointer to hover over the biquad plot point in the plot area to make the biquad filter active.

When a value for a biquad filter parameter is changed, the plot area adjusts and shows the new value in the plot.

Right-clicking a biquad filter will provide the option to copy and paste all biquad filter values.

You can set a new value with one of the methods described in the Introduction chapter, or use the biquad filter plot point as described below.

2.2.1. Filter type

The **Filter type** (2) field lets you choose a new filter type for the biquad filter from the list.

2.2.2. Center frequency

The **Center frequency** (3) field sets a new center frequency value between 10 and 20000 Hz. You can also click on the highlighted biquad filter plot point (7) in the plot area and drag the point laterally to the new value point.

2.2.3. Gain

The **Gain** (4) field lets you set a new gain value for the biquad filter. This value can be set with positive or negative values, normally ranging from -25 to 15 dB.

A. To set the value with the precision of 1 decimal point (0,1), click on the highlighted biquad filter plot point (7) in the plot area. Drag the point vertically to the new Gain point.

B. To set the Gain value manually, click the arrow up to increase the value, or the arrow down to decrease the value.

2.2.4. Q-factor

In the **Q-factor** (5) field you can set a new Q-factor value between 0,1 and 30. You can also use the mouse pointer to hover over the biquad filter plot point (7) in the plot area. Use the mouse scroll wheel to increase or decrease the value.

2.2.5. Slope

If a **Filter type** (2) has a **Slope** (6) value, the field becomes editable.

In the **Slope** (6) field you can set a new slope value for the biquad filter from the list.

NOTE: a greater slope value increases the number of biquads used.

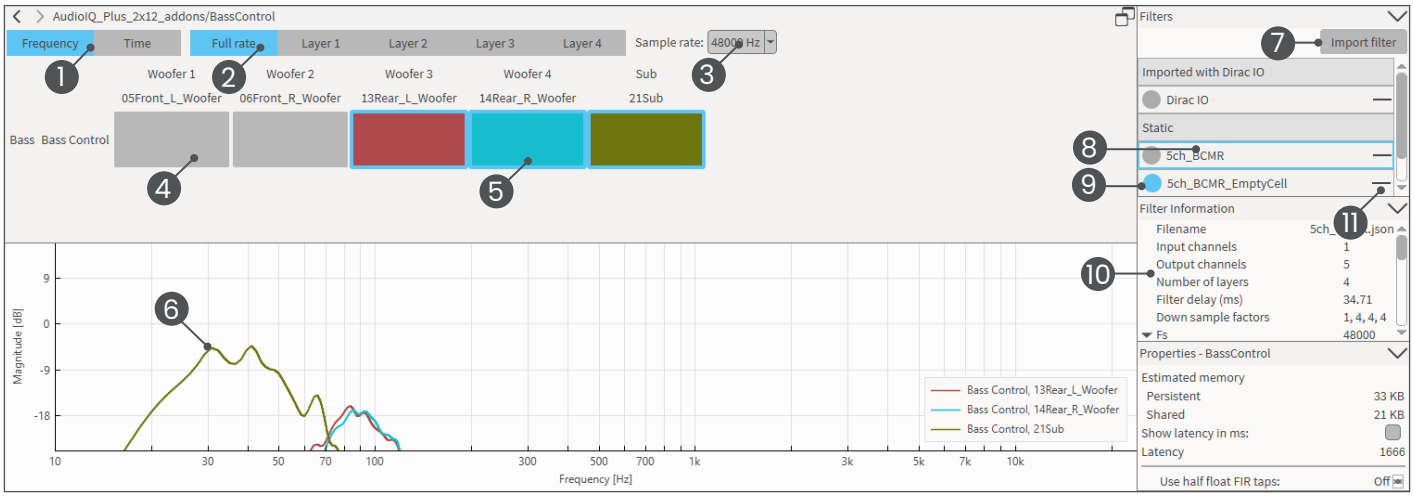
2.3. Tools

The output **Gain** (8) sets the value for the module.

The **Export** (9) tool is used when you want to export the frequency and gain values to a .txt file. This function may not be available in all workspaces.

The **Equalizer curve** tool (10) lets you use the plot area to make settings.

The **Zoom and pan** tool (11) helps you zoom in on a selection in the plot area.



1. Frequency / Time view type selection
2. Layer selection
3. Sample rate selection
4. Unselected filter cell
5. Selected filter cell
6. Plot area with selected filter cells
7. Import filter button
8. Filter shown in processor panel and properties panel
9. Filter that is sent to the device
10. Properties panel with information areas
11. Remove filter

3. Filter inspection

3.1. Overview

The **Filter inspection** processor type is where filters with set values for a number of filter cells can be imported and inspected. These processors are part of Dirac's filter technologies, for example Bass control, Harmony, Unison and others.

3.2. To import a filter

Press the **Import filter** button (7), and select a file in the correct file format for the processor (.json, XML Text, or XML Binary). You can import multiple filters into the view to inspect them one at a time.

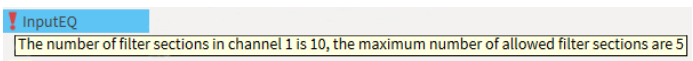
Press the file name (8) to show the filter data in the processor panel (1-6) and properties panel (10). The filter that is shown in the panels has a blue border.

To send a specific filter to the device, press the dot (9) before the filter file name. The dot becomes blue for the filter that is sent to the device.

3.2.1. Filter errors

If a filter that is sent to the device has one or more errors, the processor selection in the workspace panel shows an exclamation mark.

Put the mouse pointer over the exclamation mark to show the error information. If there are several errors, only the first is shown; when it's resolved, the next appears. Once all errors are resolved, the exclamation mark disappears and the filter can be used for the processor.



Possible filter errors can be, for example, incorrect dimensions of the filter, unsupported sample rates, incorrect filter properties, incorrect number of layers, incorrect resampling type, incorrect number of FIR taps, or too many biquads.

3.3. To inspect a filter

3.3.1. Frequency/Time

Press **Frequency** or **Time** (1) to select view type. In the plot area, **Frequency** shows the magnitude response and **Time** shows the impulse response for the selected filter cell and layer.

3.3.2. Layer

If the filter has more than one layer, you can select a layer (2) to view it in the panel. All layers are processed in parallel, where each individual layer is processed at an individual sample rate.

To view the sum of all layers, press **Full rate** in the layer selection (2).

3.3.3. Sample rate

Select the **Sample rate** (3) you want to view. You can choose between 48000 Hz and 44100 Hz.

3.3.4. Filter cells

Click on an unselected, dark grey filter cell (4) in the processor panel to select it. A selected filter cell (5) has a blue border, and the curve is shown in the plot area (6) in the the same colour as the cell colour. To show several filter cells simultaneously, press CTRL and click on the unselected filter cell you want to select and show in the plot area.

NOTE: An unused, light grey filter cell is not clickable.

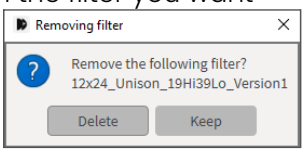
3.4. To export a filter

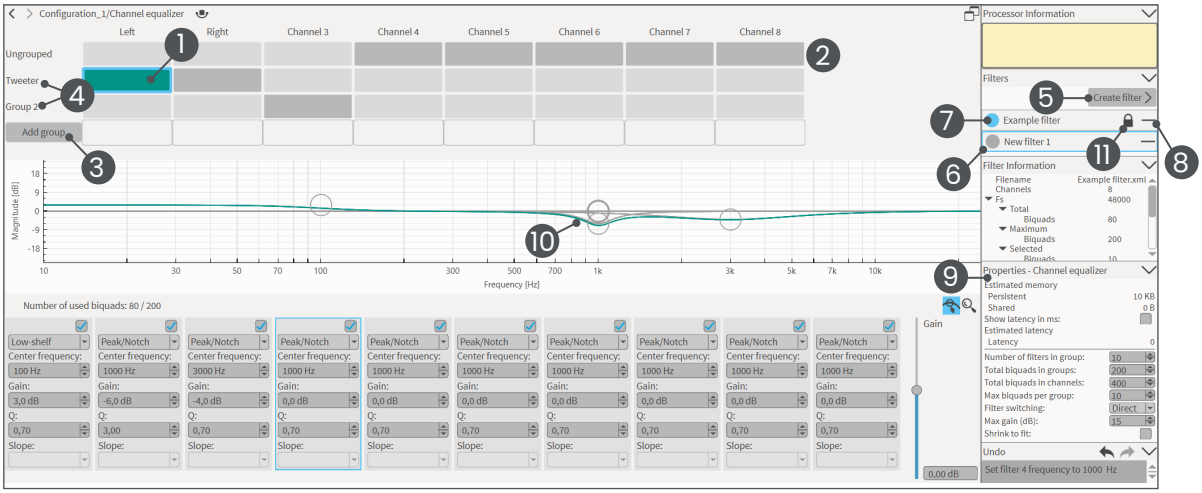
Right-click the filter you want to export (8 or 9). Select the applicable file format for this filter, and save it to your chosen location.

NOTE: Filter export requires an applicable licence.

3.5. To remove a filter

Press the minus symbol (11) on the filter you want to remove from the panels. In the popup window, press **Delete** if you want to remove it, or press **Keep** to cancel.





1. Selected filter cell
2. Ungrouped cells
3. Add group button
4. Group label
5. Create filter
6. Filter shown in processor and properties panel
7. Filter that is sent to the device
8. Remove filter
9. Properties
10. Plot area with selected filter cells
11. Locked for editing

4. Channel EQ

4.1. Overview

The **Channel EQ** is a biquad equaliser that allows you to assign different EQ settings to different sets of channels by grouping them. Each group shares a unique, editable EQ configuration. Groups are created and managed using the channel-group matrix.

4.2. To create/import a filter

Press the Create filter button (5). You can choose to:

A. Import. Select the .XML processor file you want to view and edit. You can import multiple filters into the view to inspect them one at a time.

B. Create new. Create a blank, editable filter and edit it as described below.

To import filters with Dirac IO, refer to the Dirac IO page and the Dirac IO section in the FilterLab User manual.

In the Properties panel, under the Imported with Dirac IO section or the Editable section, press the file name (6) to show the filter data in the processor panel (1-4) and properties panel (8). The filter that is shown in the panels has a blue border. To send a specific filter to the device, press the dot (7) before the filter file name. The dot becomes blue for the filter that is sent to the device.

If the imported filter has fewer channels or fewer biquad filters per group than specified in the properties, it is padded with channels and default biquad filters up to the required amount. If the filter has more channels or more biquad filters than allowed by the Channel EQ properties, only entries within the allowed range are imported. If multiple channels share identical biquad-filter sets, they are automatically grouped during import. Note that importing a filter may cause the groups to be in another order than when the filter was exported.

4.3. To edit a filter

4.3.1. Grouping channels

The channel-group matrix contains one draggable cell per column (1), representing a channel. A selected cell has a blue border, and the curve is shown in the plot area (10) in the the same colour as the cell colour.

Moving the cell to a row assigns the channel to that row's group. All the channels on the same row are in one group and share EQ settings. There can be a maximum number of groups equal to the number of channels.

When dragging and dropping a channel onto an empty row the channel retains its existing EQ settings. Dragging a channel onto a row that already contains channels causes the channel to inherit the EQ settings of the destination group. The channel's previous EQ settings are then replaced.

The top row is the ungrouped row (2). In the ungrouped row every channel is treated independently and has its own EQ settings. When dragging and dropping a channel onto the ungrouped row the channel retains its existing EQ settings.

4.3.2. Matrix rows

Pressing the "add group"-button (3) or dragging a cell to that row creates a new row. Right clicking any group label (4) will provide the option of removing the row. Any channel on the removed row will be ungrouped and retain its EQ settings. Double clicking a group label will give you the option of renaming a group.

4.3.3. To set biquad filter values

Details on configuring biquad filters are provided in the biquad-filter section of the Power EQ page.

4.3.4. Lock for editing

Filters can be locked for editing. When you are satisfied with a filter, right-click it and select Lock for

editing to prevent unintentional changes. A locked filter shows a padlock icon (11); an unlocked filter has no icon. To make the filter editable again, right-click it and select Unlock.

A filter that is imported from a file is locked by default, but it can be unlocked.

A filter that is imported with Dirac IO is always locked and cannot be unlocked, because it is controlled by the client. If you want to edit the filter, duplicate it to create an editable copy.

4.4. To duplicate a filter

If you want to make an exact copy of a filter, right-click on the filter name (6 or 7) and select **Duplicate**. The copy of the filter is shown in the parameters panel, with a number after the file name.

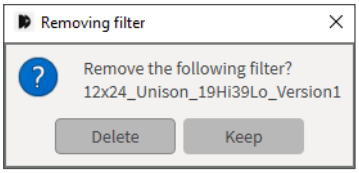
4.5. To export a filter

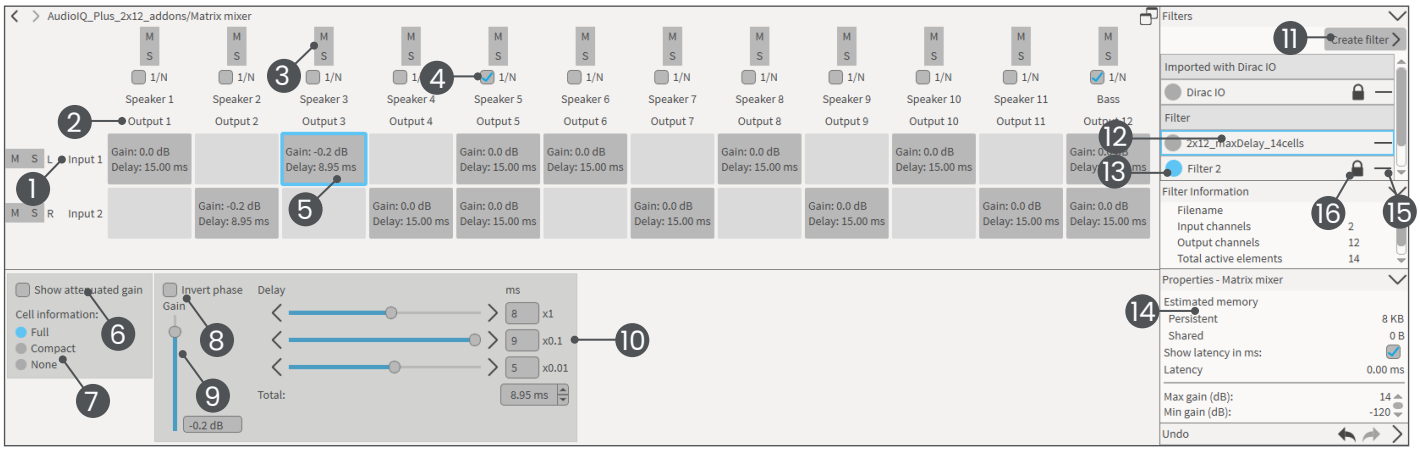
Right-click the filter you want to export (6 or 7). Select the applicable file format for this filter, and save it to your chosen location. Exported filters do not contain group names and will therefore not contain a name when imported.

NOTE: Filter export requires an applicable licence.

4.6. To remove a filter

Press the minus symbol (9) on the filter you want to remove from the panels. In the popup window, press **Delete** if you want to remove it, or press **Keep** to cancel.





1. Input channel
2. Output channel
3. M = Mute, S = Solo
4. 1/N input distribution
5. Selected filter cell
6. Show attenuated gain
7. Cell information
8. Invert phase
9. Gain
10. Delay
11. Create filter
12. Filter shown in panels
13. Filter sent to the device
14. Properties
15. Remove filter
16. Locked for editing

5. Matrix mixer

5.1. Overview

The **Matrix mixer** processor type is where each input channel can be mapped to output channels and adjusted for delay and gain. This processor type can be used for Matrix mixer, Fader, and Balance processors and others.

5.2. To create/import a filter

Press the **Create filter** button (11). You can choose to:

A. **Import**. Select the .XML processor file you want to view and edit. You can import multiple filters into the view to inspect them one at a time.

B. **Create new**. Create a blank, editable filter and add filter cells and values as described below.

To import filters with Dirac IO, refer to the Dirac IO page and the Dirac IO section in the FilterLab User manual.

In the Properties panel, under the Imported with Dirac IO section or the Editable section, press the file name (12) to show the filter data in the processor panel (1-10) and properties panel (14). The filter that is shown in the panels has a blue border.

To send a specific filter to the device, press the dot (13) before the filter file name. The dot becomes blue for the filter that is sent to the device.

5.3. To edit a filter

If you have imported a filter, or created a new filter, you can add and remove filter cells, and adjust the values for them.

5.3.1. Lock for editing

Filters can be locked for editing. When you are satisfied with a filter, right-click it and select Lock for editing to prevent unintentional changes. A locked filter shows a padlock icon (16); an unlocked filter has no icon. To make the filter editable again, right-click it and select Unlock.

A filter that is imported from a file is locked by default, but it can be unlocked.

A filter that is imported with Dirac IO is always locked and cannot be unlocked, because it is controlled by the client. If you want to edit the filter, duplicate it to create an editable copy.

5.3.2. Mute/Solo

Press **M** on the input/output channel(s) you want to mute. You can mute multiple channels.

Press **S** on the input/output channel(s) you want to listen to on their own. This mutes all other channels. You can select multiple channels as Solo channels.

5.3.3. 1/N

If there are two or more filter cells for an output, you can check the checkbox **1/N** (4) to merge the filter cell values for the output channel to prevent clipping.

5.3.4. Filter cell

Click on an unselected, dark grey filter cell (5) in the processor panel to select it. A selected filter cell has a blue border, and the cell information is shown below the matrix if it is editable.

To move a filter cell, drag it to an unused filter cell and drop it in the cell.

To add a new filter cell, right-click on an unused, light grey filter cell and select **Add element**.

To copy an existing filter cell, right-click it and select **Copy**, then right-click an unused filter cell and select **Paste**. You can also press CTRL, click the filter cell you want to copy, and drag it to an unused filter cell.

5.3.5. Show attenuated gain

Check the checkbox **Show attenuated gain** (6) to display the real gain for all filter cells.

5.3.6. Cell information

The information in the filter cells in the matrix can be adjusted under Cell information (7). To adapt to a screen resolution, it can show **Full information**, **Compact information** (values only), or **None**.

5.3.7. Invert phase

Check the **Invert phase** (8) checkbox to switch the polarity of the channel by 180°.

5.3.8. Gain

Move the **Gain** (9) fader up to increase the gain value, and down to decrease the gain value of the filter cell.

5.3.9. Delay

The **Delay** (10) to the output channel can be adjusted with high precision to a value in milliseconds.

Delay ms x1 sets the integer digits, 0-max. delay.

Delay ms x0.1 sets the first fractional digit, 0-9.

Delay ms x0.01 sets the second fractional digit, 0-9.

Total shows the combined sum of all three delay settings, for example 8.95 ms. You can also use the arrows up to increase and arrow down to decrease the delay by a full integer digit.

5.3.10. Lock for editing

Filters can be locked for editing. When you are satisfied with a filter, right-click it and select Lock for editing to prevent unintentional changes. A locked filter shows a padlock icon; (16) an unlocked filter has no icon. To make the filter editable again, right-click it and select Unlock.

A filter that is imported from a file is locked by default, but it can be unlocked.

A filter that is imported with Dirac IO is always locked and cannot be unlocked, because it is controlled by the client. If you want to edit the filter, duplicate it to create an editable copy.

5.4. To duplicate a filter

If you want to make an exact copy of a filter, right-click on the filter name (12 or 13) and select **Duplicate**. The copy of the filter is shown in the parameters panel, with a number after the file name.

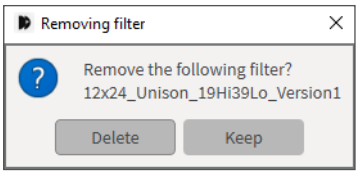
5.5. To export a filter

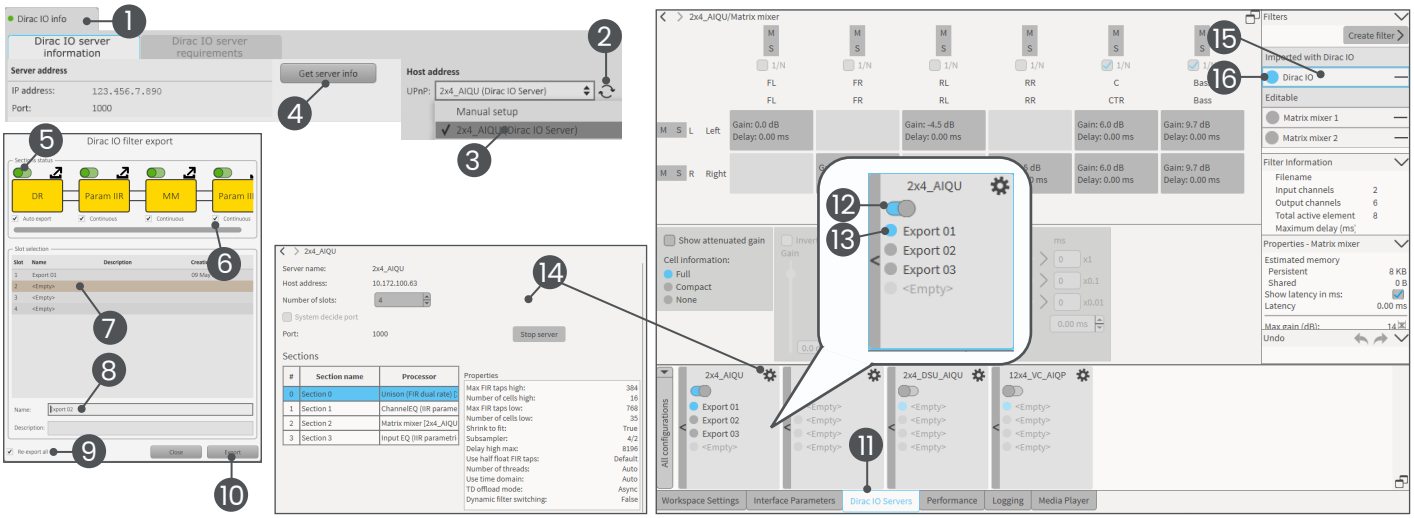
Right-click the filter you want to export (12 or 13). Select the applicable file format for this filter, and save it to your chosen location.

NOTE: Filter export requires an applicable licence.

5.6. To remove a filter

Press the minus symbol (15) on the filter you want to remove from the panels. In the popup window, press **Delete** if you want to remove it, or press **Keep** to cancel.





1. FilterLab: Dirac IO Info tab
2. FilterLab: Update UPnP icon
3. FilterLab: UPnP
4. FilterLab: Get server info
5. FilterLab export: export/exclude section
6. FilterLab export: Auto export/Continuous
7. FilterLab export: Slot selection area
8. FilterLab export: Slot name
9. FilterLab export: Re-export all
10. FilterLab export: Export
11. Flow: Dirac IO Servers tab
12. Flow: Server ON
13. Flow: Slot, enabled
14. Flow: Server properties
15. Flow: Filter shown in panels
16. Flow: Filter sent to device

6. Dirac IO

6.1. Overview

The **Dirac IO** makes it possible to create filters in FilterLab and import them in Flow to display them in real time in specified processors.

6.2. To import a filter with Dirac IO

When filters have been created in FilterLab, they can be exported from FilterLab, imported into Flow with the Dirac IO connection.

The following instructions are for an existing server connection between FilterLab and Flow. For instructions on server implementation, refer to the section on Dirac IO integration in the FilterLab User manual.

6.2.1. Set the server in Flow

1. Go to the Settings panel and press the **Dirac IO Servers** (11) tab.
2. Set the server you want to use to ON (12).

6.2.2. Set the server in FilterLab

This must be set the first time the Dirac IO connection is made. Depending on setup, the setting can be saved.

1. Press the **Dirac IO info** (1) tab.
2. Press the update UPnP icon (2).
3. In the **UPnP** field (3), select from the list the name of the server you set to ON in Flow.
4. Press the button **Get server info** (4).
5. Make sure that there are no errors or warnings in the server connection. An error is shown with a red icon you can click on to get more information.



6.2.3. To do a Dirac IO export in FilterLab

1. In the top menu, go to Filter export/Dirac IO export.
2. In the popup, select/deselect the sections you want to export/exclude (5), and check the checkbox (6) below the section if you want an automated or continuous feed of data from the sections in FilterLab.

NOTE: In the export popup, a section is yellow if it is exported for the first time, if there are changes made to the section since last export, or if a different slot than the previous export is selected.

A section is green if your previous export was sent to the same slot and there are no changes made to the section.

3. In the Slot selection area (7), select the server slot that you want to send the export to. You can choose to give the selected slot a new name (8).

NOTE: A slot contains the data for each of the selected sections of the export.

Example: To do A/B testing you can export version A for each of the sections to slot 1, and version B of each of the sections to slot 2. You can then quickly switch between slot 1 and slot 2 in Flow to test the different versions.

4. Check the checkbox **Re-export all** (9) if you want all the filter data exported in full to Flow.
5. Press **Export** (10) to send the filter through Dirac IO.

NOTE: An export *overwrites* any previous export to the selected slot.

6.2.4. To do a Dirac IO import in Flow

1. Go to the Dirac IO Servers (11) tab in the Settings panel.
2. Press the dot (13) by the slot name you sent your export to. An enabled slot has a blue dot.
3. Press the cogwheel of the server (14) to view the server properties, imported sections and their parameters. You can also adjust the number of slots or stop the server.
4. Go to the properties panel and press the Dirac IO filter name (15) to show it in the processor and parameter panels. The filter must have a blue border.
5. Press the dot (16) in front of the Dirac IO filter to send the filter to the device. The dot becomes blue for the filter that is sent.

6.2.5. To show filter adjustments in real time

To show changes made in FilterLab in real time in Flow, make sure these parameters are set:

In FilterLab:

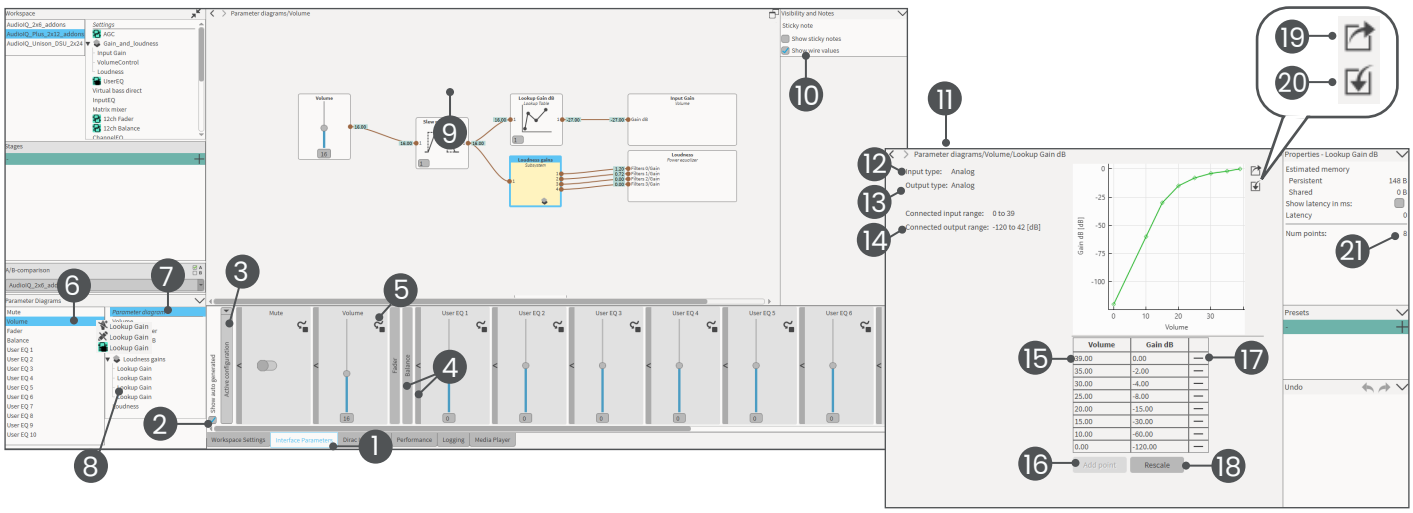
1. Set Auto export/Continuous (6) for each section in the export to a selected slot in Flow.

In Flow:

2. Enable the correct slot (13) on the server on the Dirac IO Servers tab.
3. Click the Dirac IO filter name (15) in the parameters panel to show the filter in the processor panel. The Dirac IO filter must have a blue border.
4. Press the dot (16) in front of the Dirac IO filter name. The dot becomes blue for the filter that is sent to the device.

6.3. To edit a filter

A filter imported with Dirac IO to Flow is in most cases editable only in FilterLab.



1. Interface parameters tab
2. Show auto generated
3. Parameter view
4. Expand/collapse parameter
5. Shortcut to parameter diagram
6. Interface parameter view selection
7. Parameter diagram view
8. Parameter proessor list
9. Parameter processor
10. Show wire values
11. Lookup table settings view
12. Input type
13. Output type
14. Input/output range
15. Filter shown in panels
16. Add point button
17. Remove point icon
18. Rescale button
19. Export table
20. Import table
21. Number of points

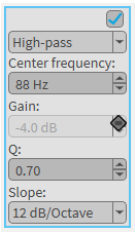
7. Interface parameters

7.1. Overview

The **Interface parameters** are selected parameters that can be changed by external input, for example a user interface. The change triggers a re-calculation of the parameter. The new calculation value is sent to the signal chain that executes the change in the output.

The parameters on the Interface parameters tab (1) in the settings panel are the parameters that can be set up to be changed from the device user interface.

7.2. Parameters



When a signal chain processor has a parameter that can be changed externally, the processor shows the parameter field as locked for editing with a lock icon ◆ next to the field in the processor panel.

The parameters can be configured as required.

7.2.1. To view interface parameters

1. Press the Interface parameters tab (1) in the settings panel to view the Interface parameters in the workspace.
2. If you want to see the automatically generated parameters, check the checkbox **Show auto generated** (2).
3. Select parameter view (3). You can choose to show **All configurations** (all parameters in all configurations), **Active configuration** (relevant parameters for the selected configuration), or a specific configuration.
4. If you want to optimise the view, you can click on the arrow icon (4) to minimise a parameter in the view, or click on the parameter name to expand it to normal view.

7.2.2. To view interface parameter diagrams

Interface parameters are defined in the interface parameter diagrams.
NOTE: An interface parameter diagram can contain more than one interface parameter.

1. To view the parameter diagram of an interface parameter and its processors:
   A. Click on the shortcut (5) on the interface parameter to go directly to the diagram.
   B. Click on the parameter diagram (6) you want to show. If necessary, select the Parameter diagram view (7).
2. To show the values that are sent from the parameter through the processors to the signal chain processors, check the checkbox **Show wire values** (10).
3. Double-click on a processor (9) or click on a processor name in the parameter processor list (8) to go to the processor settings view (11).

If a processor in the Parameter diagram cannot be double-clicked, it does not have a view.

7.3. To adjust an interface parameter

7.3.1. Interface parameters panel

An interface parameter is shown as a slider or a switch. The slider can be moved up to increase the value or down to decrease the value of the parameter. The switch is an On/Off button that has two positions. The change to a parameter in the panel is the same as the change an end user makes on the user interface.

7.4. To adjust processor settings

7.4.1. Processors

The settings of an interface parameter diagram processor are in some cases editable.

Go to a processor settings view (11) as described above. If the processor settings are editable, you

can adjust the values in the view.

Some of the settings are view-only that for example show the processor name, parameter ID, type, or the mapping to the signal chain processor.

There are different types of processors that have different settings available, for example Slew rate limiter, On/off, and Lookup table.

If there is an icon before a lookup table processor in the parameter processor list (8), the lookup table processor can be:


Restricted for editing during tuning

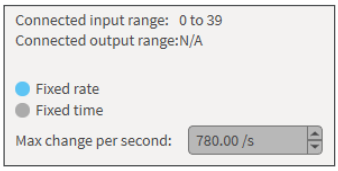

Not editable during tuning


Shared, can be used in multiple interface parameter diagrams

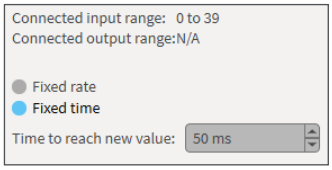
7.4.2. Slew rate limiter

When a user uses an interface parameter to increase or decrease the output, the Slew rate limiter can be set to limit the rate or time between the old value and the new value:

A. **Fixed rate**. It sets a maximum rate change per second.

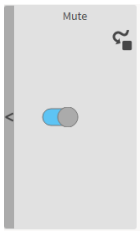


B. **Fixed time**. It sets the time to reach the new value.



7.4.3. On/off

The On/off parameter has two settings: **On** or **Off**, where one of the values can be set as default. One application area of this processor type is Mute.



7.4.4. Lookup table

A lookup table processor for an interface parameter can give more control over how the parameter value change is calculated before it is sent to the signal chain processor for output.

You can see the **Input type** (12) and **Output type** (13) of the lookup table in the processor panel.

The lookup table processor settings view also shows the ranges of the input and output (14).

A lookup table has a set number of points (15). The table can use some of the points, or all of them. The amount of points can be seen under **Num points** (21) in the properties panel.

1. Click the **Add point** (16) button to add another point within the set number of points. If the button is grey and not clickable, the maximum number of points has been set for the table.
2. Click the **Remove point** (17) icon on a point to remove it.
3. Click the **Rescale** (18) button to set new minimum and maximum values for the lookup table. In the popup window, set the new values and click **Rescale** to recalculate the table, or **Cancel** to discard the changes.
   NOTE: Rescale lets you set the table values outside the connected input and output range.

To change the values in a lookup table

To change some of the values of a lookup table, you can click in the field you want to change and type a new value.

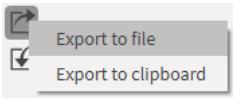
You can also use the arrow up to increase or arrow down to decrease the value by a set increment.

Alternatively, you can click on the value and use the mouse scroll wheel (if enabled) and scroll up to increase or down to decrease the value.

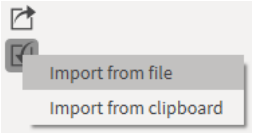
To export and import a lookup table file

If a lookup table has the correct values and number of points you want to use in another lookup table processor, you can export the lookup table so that it can be imported to another table.

1. Click the **Export table** (19) icon.
2. Select **Export to file** to export the table as a .paramMap file. A file window opens.



3. Set the file name and location where the file is to be saved.
4. Go to the lookup table processor you want to import it to.
5. Click the **Import table** (20) icon and select **Import from file** to import a lookup table file to a lookup table processor.



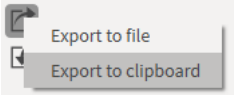
6. Select the correct .paramMap file or a correctly structured .csv file in the file window.
7. Press **Open** to import the file to the lookup table processor.

NOTE: If the import is made correctly, the imported values overwrite any existing values and are shown in the lookup table processor. If there are errors in the import, an error prompt is shown with more information. Correct the error, and do a new import.

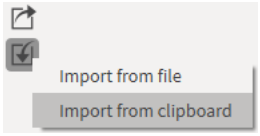
If the number of points of the import is different from what the original export was, the new number of points is displayed if it is within the set amount of points.

To export and import with the computer clipboard
To change several values of a lookup table processor quickly and easily, you can export and import the lookup table with the help of the computer clipboard.

1. Click on the **Export table** (19) icon and select **Export to clipboard**.

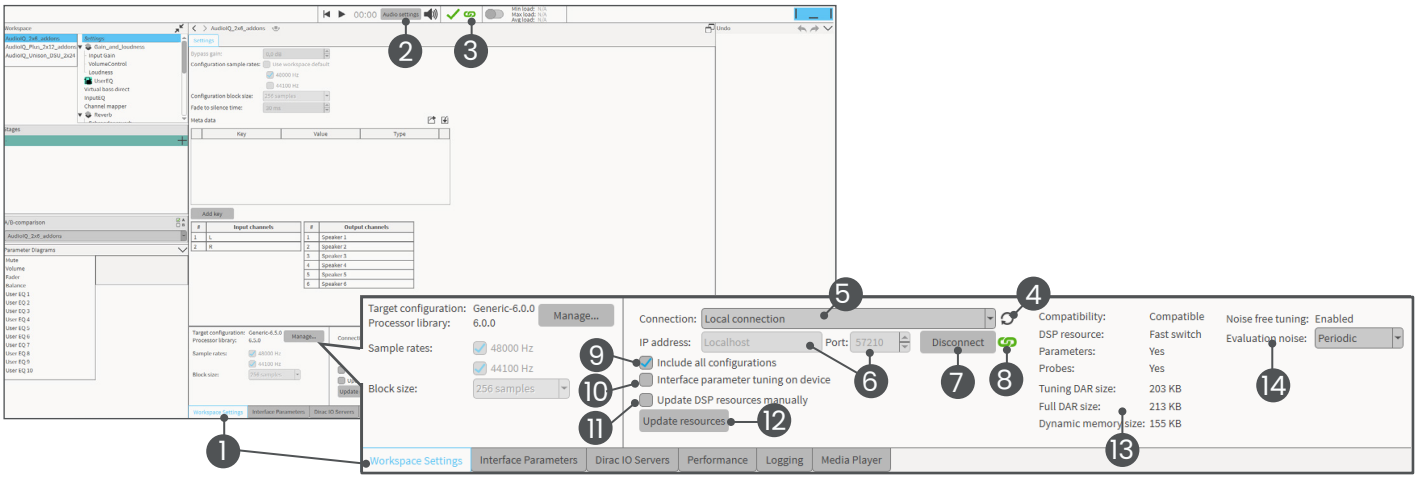


2. Open Excel, click on an empty cell and press CTRL+V or right-click and select Paste to add the content of the clipboard to a spreadsheet. You can also open Notepad or Word and paste the content into the file.
3. Do the changes to the values you want to adjust.
4. Select the table cells/content in Excel/Notepad/Word and press CTRL+C or right-click and select Copy.
5. Go to Flow, press **Import table** (20) icon and select Import from clipboard.



NOTE: If the import is made correctly, the imported values overwrite any existing values and are shown in the lookup table processor. If there are errors in the import, an error prompt is shown with more information. Correct the error, and do a new import.

If the number of points of the import is different from what the original export was, the new number of points is displayed if it is within the set amount of points.



1. Workspace Settings tab
2. Audio Settings button
3. Connection status display, screen
4. Update available connections
5. Connection selection
6. IP address and Port
7. Connect/Disconnect button
8. Connection status display, tab
9. Include all configurations
10. Interface parameter tuning on device
11. Update DSP resources manually
12. Update resources button
13. Connection data
14. Noise settings

8. Device connection

8.1. Overview

The **Workspace Settings** tab (1) has the device connection settings that are used for setting up the the connection from Flow to a device for tuning.

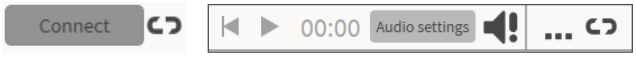
8.2. To connect

To connect Flow to the device:

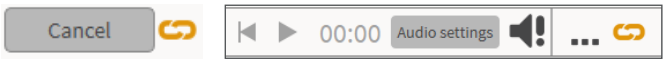
1. Press the **Workspace Settings** (1) tab in the settings panel.
2. If you process audio on your computer or stream audio from the Flow Media player, click on the **Audio Settings** (2) button to open the Audio settings application. When connected, the loudspeaker icon is shown as at the top of the screen.
3. Press the Update available connections (4).
4. In the **Connection** field (5), select from the list the type of connection you require:
   - A. **Local**. The setup connects locally on your computer only, and the IP address and port are set automatically.
   - B. **Manual**. You must type in the IP address and Port (6) of the device manually.
   - C. Specific network-connected device that is shown in the list. The IP address and port are set automatically.
5. Press the **Connect** (7) button to connect to the device.

The connection status display (8) is shown next to the **Connect** (7) button on the Workspace Settings tab, and at the top of the screen (3).

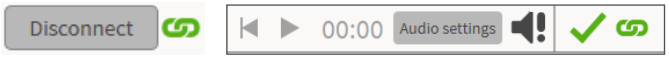
When Flow is not connected to a device, the connection status display is shown as disconnected.



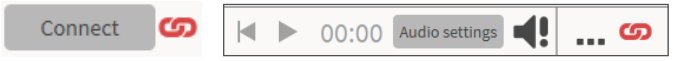
When the connection is in progress, the connection status changes shape and the display is yellow:



If the connection is made correctly, the connection status display (3, 8) changes shape again and is marked in green.



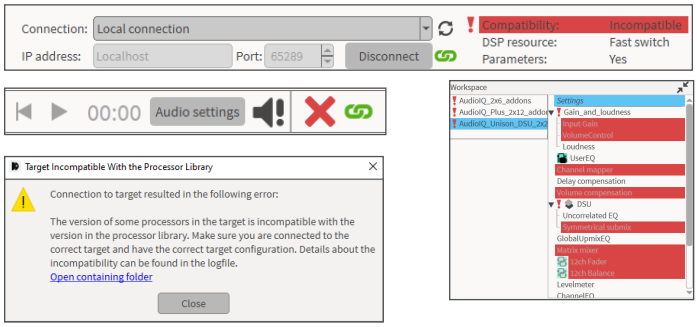
If the connection failed, the connection status display (3, 8) is shown as red.



To get more information on any errors, start logging the device connection. For instructions, refer to the section below on logging.

If the connection was made correctly, but the target configuration is incompatible with the processor library, you may have a version of the processor library and a version of the RT Core of the device that cannot function correctly together for all processors.

The connection status display is shown as green, but the Compatibility connection data and the incompatible processors are marked in red. A popup window is shown with more information.



Make sure that the target configuration file, the processor library and the device runtime are compatible.

8.3. Connection parameters

8.3.1. Include all configurations

The checkbox **Include all configurations** (9) is checked by default, and lets you send all the configurations of your Workspace to the connected device. If the checkbox is not checked, only the selected configuration is sent to the device.

8.3.2. Interface parameter tuning on device

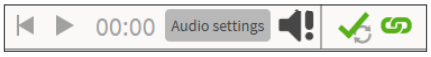
Check the checkbox **Interface parameter tuning on device** (10) to make the calculations for the interface parameter directly on the device.

8.3.3. Update DSP resources manually

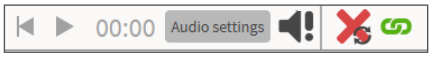
Check the checkbox **Update DSP resources manually** (11) if you want to manually update the resources sent between the device, Audio Settings and Flow. This can be useful if you have a very large workspace and work on a slow connection.

The automatic transfer of signal chain data is stopped, and the resources are updated only when you press the button **Update resources** (12).

In manual update mode, the signal chain icon has an update icon that is green when the signal chain is up-to-date in the ACS (audio control service) for the connected plugins, devices and Flow.



If the signal chain icon is red, the signal chain is not synchronized with the ACS and you must press the **Update resources** (12) button.



8.3.4. Connection data

The connection data (13) of the Workspace Settings tab shows information about the connection and the files sent to the device.

Compatibility shows if the target configuration and processor library and compatible or incompatible. For more information, refer to the section above on how to connect.

DSP resource shows the speed of how fast the response is from the device for the DAR files that are sent to the device from Flow.

Parameters shows if the connected device supports tuning using module parameters. If the connected device does not support module parameters, changes that are made in Flow can only be updated by sending a new DAR file to the device.

Probes shows if the connected device supports information about instantaneous signal levels at points in the signal chain. These values are often shown in Processor views that use level meters.

Tuning DAR size shows the size of the DAR file used during tuning.

Full DAR size shows the size of the full DAR file.

Dynamic memory size shows how much device memory is used for processing.

8.3.5. Noise

The Noise settings (14) of the Workspace Settings tab shows information about evaluation noise.

Noise-free tuning shows if tuning without evaluation noise is enabled, disabled, or pending (waiting for data). The evaluation noise event will be checked against the evaluation license key at intervals to verify if the license is valid and the tuning can be made noise-free.

Evaluation noise sets the interval of the evaluation noise that is sent to the device. The evaluation noise can be set to **Periodic** (regular intervals) or **Randomized** (irregular intervals within a longer time span).

8.4. Performance measurement

The **Performance** tab in the Settings panel in Flow has settings for the performance measurement of the processors. This is useful for an analysis of the connection and possible performance congestions.

When performance measurement is set to On, Flow shows the Min load, Max load and Avg load for the entire connection at the top of the screen next to the connection status display (3).



To view the performance of a specific processor, click on the processor in the workspace panel. A box with the performance values is shown on the **Performance** tab.

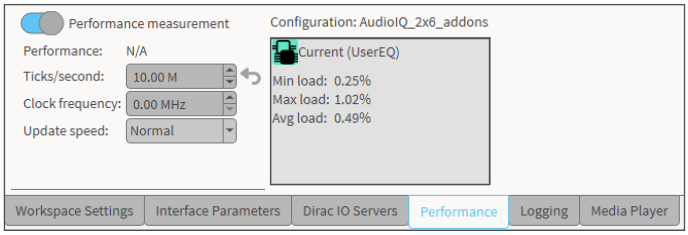
The performance measurement can be made more precise with these settings:

Ticks/second. Sets the amount of ticks per second. The value must correspond to the actual amount of ticks that are possible for the ACS.

Clock frequency. Sets the clock frequency for the ACS. When the correct clock frequency is set, the performance measurement values at the top of the screen are also shown in MCPS (mega cycles per second).



Update speed sets the speed of the performance measurement. If you work on a slow connection, set the update speed to slow.



8.5. Logging

Flow logs events continuously. The Logging tab in the Settings panel shows the log messages, and is useful for troubleshooting a connection to identify issues and get more information about them.

Set **Logging on device** to On to also log events on the device itself. This gives you control over the resources used for logging, because you can filter what is logged on the device before the data is sent to Flow, rather than filtering only in the list.

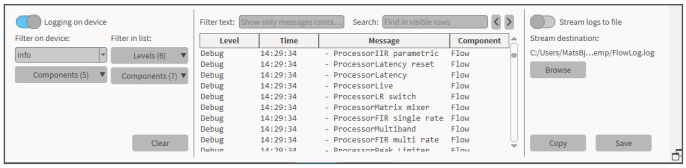
Filter on device lets you choose the log level and the components to log on the device. **Filter in list** lets you filter the levels and components that are shown in the log message list in Flow.

Use the Filter text field to show only messages that contain specified text, and use Search to find and step through matching rows with the arrow buttons

Press **Clear** to clear the log message list.

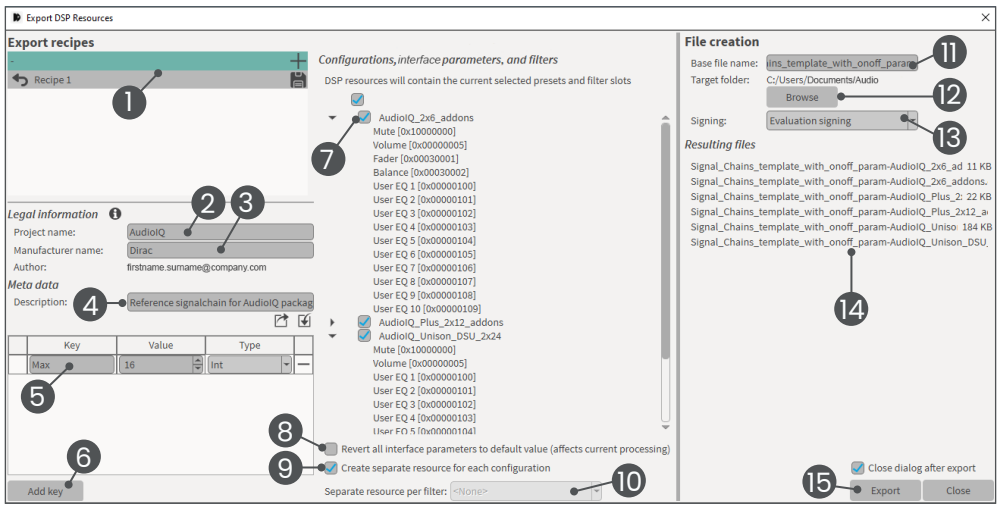
Turn on **Stream logs to file** to continuously write the log to a file on your computer, and press **Browse** to set the Stream destination. This lets you open and inspect the log in the text editor of your choice.

Press **Copy** to copy the log messaes to the cliboard, or press **Save** to save the log to a file on your computer.



8.6. To disconnect

If you want to stop an existing connection, click the **Disconnect** (7) button. This stops the connection, and the connection status display is set to disconnected.



1. Export recipe
2. Project name
3. Manufacturer name
4. Meta data description
5. Meta data line
6. Add key button
7. Configurations, Interface parameters, filters
8. Revert all interface parameters to default value
9. Create separate resource for each configuration
10. Separate resources per filter
11. Base filter name
12. Target folder
13. Signing
14. Resulting files
15. Export button

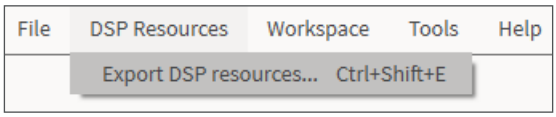
9. DSP Resources export

9.1. Overview

The **DSP Resources export** is where configurations and interface parameters can be exported to be transferred to a device for tuning or installation.

9.2. To export DSP resources

Save your current workspace. Then go to the top menu to DSP Resources/Export DSP resources. A separate window opens.



9.3. To do a quick export

In the Export DSP Resources window:

1. In the Legal information area, set the **Project name** (2) and **Manufacturer name** (3) that Flow is licensed to. Refer to the section on Legal information below for more information.
2. Check the checkbox for each configuration that you want to export as files.
3. If you want to create separate files for each configuration, check the checkbox **Create separate resource for each configuration** (9).
4. Select **Target folder** (12) on your computer where you want to save the exported files.
5. Set the **Signing** (13). Refer to the section on File creation below for more information.
6. Make sure the **Resulting files** (14) are as you set them (one large file / one file per configuration). NOTE: The export file pairs must be kept together, as both files are required for successfully running the files on the device.
7. Press the **Export** button (15) to export the files.

9.4. To do an export with specific settings

If you want to do an export with more information than a quick export, you can export with specific settings and save the export settings as an export recipe.

9.4.1. Legal information

The text in the Legal information area is used for legal information connected to software licensing.

Project name (2) refers to, for example, the name of the product model that is on the Flow software license.

Manufacturer name (3) refers to the company name that Flow is licensed to.

Author. The user that is logged in is set automatically as the author of the files.

9.4.2. Meta data

Meta data are features with values that are applied in a specified way in an integration with a device. Meta data is project-specific, and requires a customised setup. To use meta data for your project, speak to Dirac.

9.4.3. Configurations, interface parameters, and filters

The configuration that is selected in Flow is also selected automatically for export.

If you want to include all workspace configurations in the export, check the checkbox at the top of the list of configurations to select all.

If you want to select specific configurations, check the checkbox (7) for each of the configurations you want to include in the export.

If you want to set all interface parameters to their default value for the export, check the checkbox **Revert all interface parameters to default value** (8).

NOTE: this setting has a direct effect and will set all interface parameters to default in the ongoing processing.

If you want to have each configuration as a separate file in the export, check the checkbox **Create separate resource for each configuration** (9).

If a Filter inspection processor has more than one filter, you can select to create a separate file for each filter. Select the filter processor from the drop-down list in the **Separate resource per filter** (10) field.

NOTE: you cannot check the checkbox **Create separate resource for each configuration** (9) and select **Separate resource per filter** (10) simultaneously.

9.4.4. File creation

Base file name. If you want to change the name part of the export files, you can edit the name in the **Base file name** (11) field. The export file names are set with the edited Base file name as the new file name.

Target folder. The system sets automatically a folder where the files will be saved. To set a different folder for the exported files, press the **Browse** (12) button and select a new folder in the window. Press Select folder to confirm.

Signing. The **Signing** (13) of the files is preset to **Evaluation signing.** This license type is available for tuning the device, and the files have an evaluation noise.

For production/implementation files that do not have an evaluation noise, speak to Dirac.

9.4.5. Resulting files

The Resulting files (14) area shows the files that are generated during the export. An export generates a .DAR file and a .json file, and the .DAR file value shows the calculated file size of the exported file.

If the checkbox **Create separate resource for each configuration** (9) is checked, the Resulting files area shows a .DAR file and a .json file for each configuration.

The configuration(s) are exported as .DAR file(s) that contains meta information about the configuration and the interface parameter(s). Additionally, the setup and interface parameters are exported as .json file(s).

The export file .DAR and .json pairs must be kept together, as both files are required for successfully running the files on the device.

NOTE: If you have already exported the files before to the same folder, the system sets a warning icon ⚠ in front of the names of the resulting files. The files that have the the warning icon are overwritten by the new export.

9.4.6. Export recipes

When you have made all the specific the settings you want for an export, you can save the settings as an Export recipe (1). Click on the + icon to save the current setting as an export recipe. You can edit the name of the export recipe to give it a specific name.

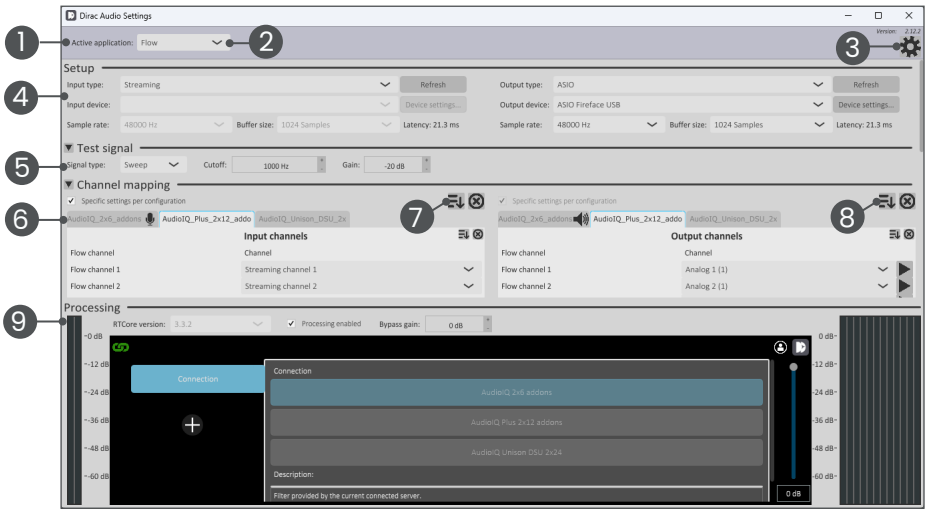
The export recipes are saved in your workspace and can be used when required.

9.4.7. Export

If you want to close the Export DSP Resources window after the export is done, check the checkbox **Close window after export**.

To export the files to the selected folder, press the **Export** (15) button.

If you want to close the Export DSP Resources window without exporting, press the **Close** button.



1. Application settings panel
2. Active application
3. Settings
4. Setup panel
5. Test signal panel
6. Channel mapping panel
7. Automatic channel mapping, input
8. Automatic channel mapping, output
9. Processing panel

10. Audio settings

10.1. Overview

In Audio Settings, you do the required settings to map the input and output channels for the device when you process audio on your computer or stream audio from the Flow Media player.

10.2. Audio settings panels

10.2.1. Application settings panel

In the Application settings panel (1), set Active application (2) if required.

The panel also has the Settings (3) cogwheel to select **Theme** (dark/light), **Save settings as**, and **Load settings**.

If you have saved audio settings available, you can load the settings into Audio Settings. Press the **Settings** (3) cogwheel and select **Load settings**. Choose the correct settings .json file and press Open.

If you want to save your settings, press the **Settings** (3) cogwheel and select **Save settings as**. Choose a name for the settings file, select a location, and press Save.

10.2.2. Setup panel

In the Setup panel (4), you can on the input side set the **Input type**. If applicable, press the **Refresh** button to update the list of input devices of the selected input type. If the input type and device have any of these settings, you can also set: **Input device**, **Sample rate**, **Buffer size**.

On the output side set the **Output type.** If applicable, press the **Refresh** button to update the list of output devices of the selected output type. If the output type and device have any of these settings, you can also set: **Output device**, and if applicable: **Sample rate**, **Buffer size**.

When you are streaming an output, you must set **Host**, **Port**, and press **Connect** to set up a streaming connection.

10.2.3. Test signal panel

On the Test signal panel (5), you can set the **Signal type** (Sweep/Pink noise) for the test signal, and increase or decrease **Cutoff** and **Gain**.

10.2.4. Channel mapping panel

On the Channel mapping panel (6), the input channels are mapped to the input channels in Flow, and the output channels in Flow are mapped to the device output channels.

The Input channels area shows the different configurations from Flow as tabs, and the configuration that is selected in Flow has a microphone icon on the tab.

Map each Flow input channel to an input channel from the input device. You can also set the first one and then press the **Automatic channel mapping** (7) icon to let the application map the other input channels. Each input channel can only be mapped once per configuration.

If you want to make different input channel settings for each configuration, check the checkbox **Specific settings per configuration**.

The Output channels area shows the different configurations from Flow as tabs, and the configuration that is selected in Flow has a speaker icon on the tab. Map each Flow output channel to each of the device output channels.

You can also press the **Automatic channel mapping** (8) icon on the Output side to let the application map the channels. Each output channel can only be mapped once. If you want to make different output channel settings for each configuration, check the checkbox **Specific settings per configuration**.

10.2.5. Processing panel

The Processing panel (9) shows the RT Core version, the connection status, and the selected configuration that is processed. It also makes it possible to adjust the Bypass gain and Overall gain.

Check the checkbox Processing enabled to show the processing in the panel. If you uncheck the checkbox, the processing stops.

If you click on the profile icon you can sign out of the processing part of Audio Settings.

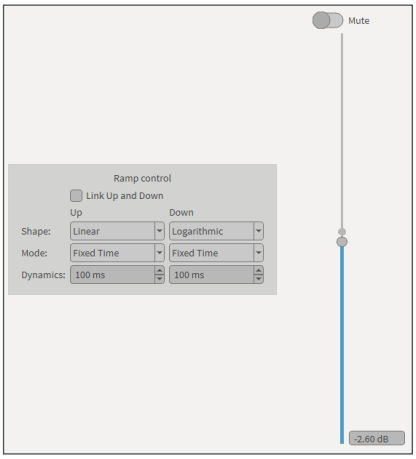
11. Utility modules

11.1. Overview

The utility modules described below are modules that help you inspect, map, view, edit, and play audio.

11.2. Volume

Volume is where gain can be applied to a signal or the signal chain.



The Ramp control area has settings for the ramp up and ramp down:

Link up and down lets you set the same ramp control values for both up and down in one place.

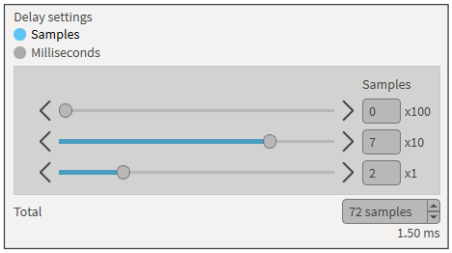
Shape sets if the shape is **Linear** or **Logarithmic**.

Mode sets if the values are shown as **Fixed Rate** (g/s), **Fixed time** (ms) or **One block**.

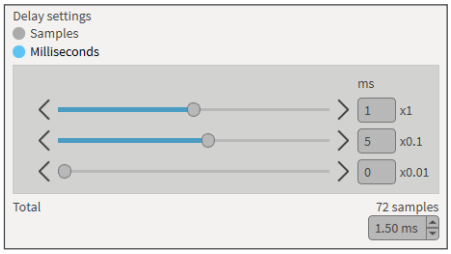
Dynamics shows the value and lets you to edit if the mode is Fixed rate or Fixed time.

11.3. Delay

The Delay module lets you set the delay, and you can choose between two different units: Samples or Milliseconds. Press the dot in front of the unit to select the unit for the settings and the view.



Delay display in Samples



Delay display in Milliseconds

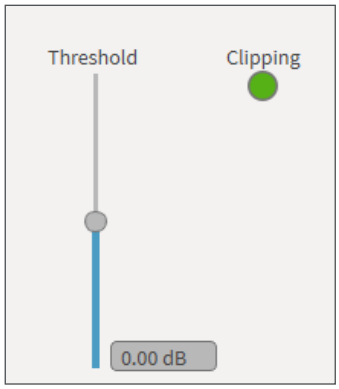
11.4. Channel Mapper

Channel mapper lets you map the input channels to the output channels. Click on an element to enable (blue) or disable (grey) it.

The module has a Compact and a Full view mode. In Full view mode, the channel names are shown. In Compact view mode, only the channel index is shown.

The module also lets you create a diagonal matrix. This clears any previous selection.

When you set two or more filter cells for an output channel, the gain automatically merges the filter cell values for the output channel to prevent clipping. It follows the same equation as the 1/N in the Matrix mixer. Refer to the Matrix mixer page for more information.



11.5. Clipper

The output of the Clipper module is clipped at the threshold value. Below this value, audio will go through unaffected.

When the Clipping dot is red, it shows that the threshold has been breached and clipping occurs.

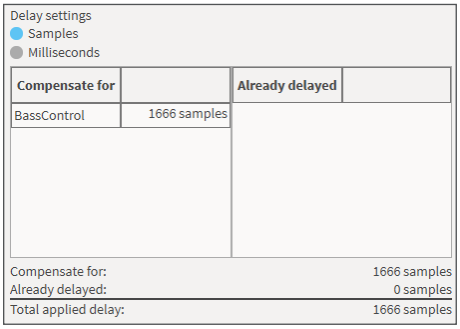
When the audio is within a range that does not cause clipping, the Clipping dot is green.

To prevent clipping,
adjust the relevant values until there is no more clipping.

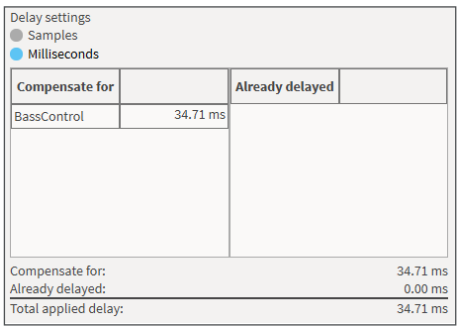
11.6. Delay compensation

The Delay compensation module is for inspection of the delay values for defined processors. The delay applied is the sum of the values in the “Compensate for” column minus the sum of the values in the “Already delayed” column.

The delay information can be inspected in two different units. Press the dot in front of the unit to select the unit for the view.



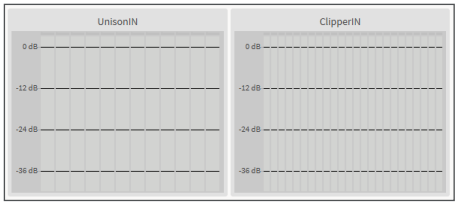
Inspection display in Samples.



Inspection display in Milliseconds

11.7. Custom Panel

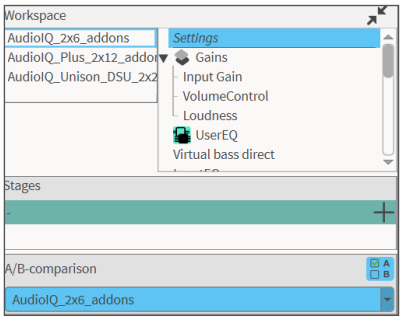
It is possible to have a customised panel in Flow that has a display for specified levels or parameters. This panel is set up by Dirac on request.



11.8. A/B Comparison

If you want to switch between two configurations quickly with one click, you can use A/B-comparison.

1. Select the first configuration you want to include in the comparison in upper left corner of the workspace panel.
2. Go to the A/B-comparison area in the middle of the workspace panel and select from the drop-down list the second configuration you want to include in the comparison.
3. To switch between the two selected configurations, press the A/B button to jump from one to the other. You can also press Shift+F12 to toggle between the two configurations. The configuration that is active is shown as blue.



11.9. Media Player

The media player is located on the Media Player tab in the Settings panel. If you use Audio Settings for audio processing, set the input type to Streaming to enable playback of an audio file in Media Player.

11.9.1. To open a media file

Press the **Open audio file** (1) button to select a file from your computer to upload as a track. The accepted file formats are .wav / .mp3 / .flac / .ogg.

11.9.2. Playback

If you want to adjust the compression of the track for playback, select the compression level in the drop-down list of the **Compression** (2) field.

Press **Play** (7) to start the playback of the track with the settings you made. If you have checked the checkbox **Loop** (4), the playback will play within the loop selection (11).

Press **Back to start** (6) to go to the beginning of the track.

11.9.3. Loop

If you want to loop a specific section of the track, use the two dark grey Loop markers (10) to define the loop selection (11). The grey area is the selection that is played in a loop.

Check the checkbox **Loop** (4). Press **Play** (7) to start the playback. The track now loops infinitely in the defined selection.

11.9.4. Playback position marker

The blue **Playback position marker** (8) displays the playback position on the track plot.

To zoom in on a part of the track, hover over the track plot with the mouse pointer and use the mouse scroll wheel to zoom in.

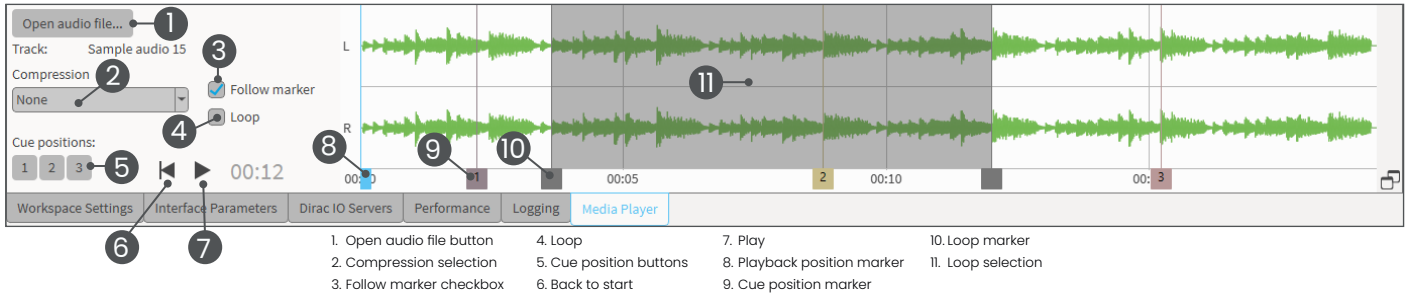
If you want to follow the playback when the plot area is zoomed in, check the checkbox **Follow marker** (3) to show the position of the **Playback position marker** (8) when it moves through the track.

11.9.5. Cue position markers

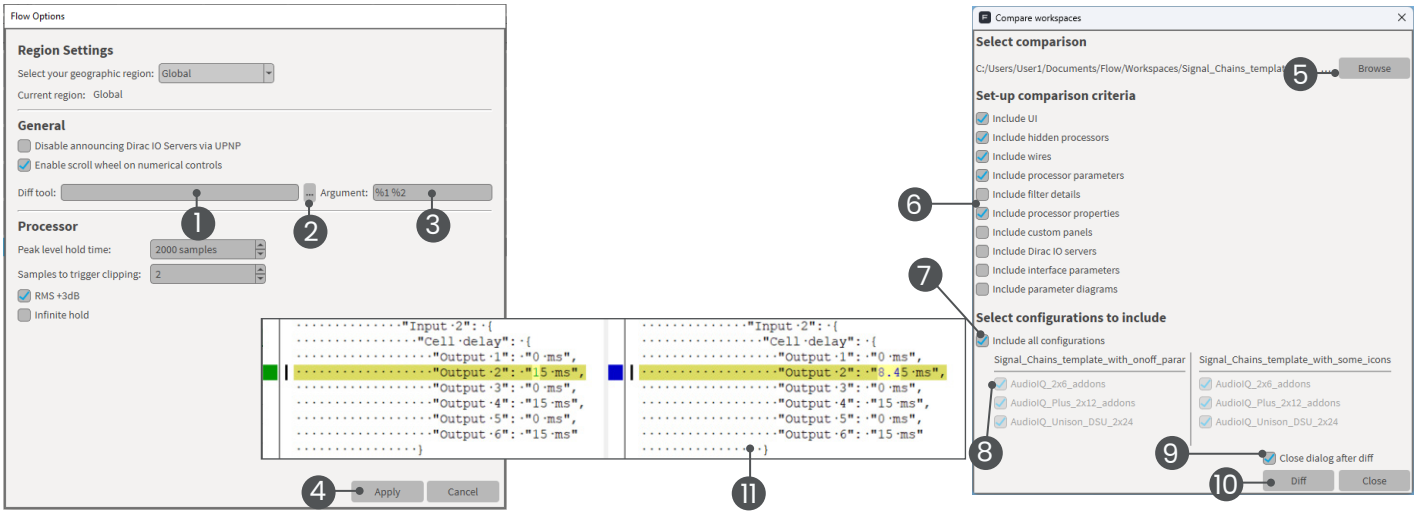
You can set 3 **Cue position markers** (9) on the track. Drag a cue position with your mouse pointer to the position you want to have it in.

When you have set each of the cue position markers, press the **Cue position** (5) number button to jump to the selected cue position.

Media Player



1. Open audio file button
2. Compression selection
3. Follow marker checkbox
4. Loop
5. Cue position buttons
6. Back to start
7. Play
8. Playback position marker
9. Cue position marker
10. Loop marker
11. Loop selection



1. Options: Diff tool field
2. Options: Select diff tool file
3. Options: Argument for Diff tool
4. Options: Apply
5. Advanced compare: Browse
6. Advanced compare: Comparison content selection
7. Advanced compare: Include all configurations
8. Advanced compare: Select configurations
9. Advanced compare: Close dialog after diff
10. Advanced compare: Diff
11. Diff tool: Comparison

12. Compare workspaces

12.1. Overview

The **compare workspaces** feature is a fast way to analyse the differences between two workspaces. All parameters and their values in the two workspaces are exported to a Diff tool, a difference comparison application, where the differences between the files are highlighted.

12.2. Diff tool

To be able to use the comparison feature in Flow, you must have a diff tool installed locally on your computer. The application must be able to compare .json files.

To set up the diff tool in Flow:

1. Go to the top menu Tools / Options.
2. In the General area, press the dots (2) next to the Diff tool field (1).
3. In the window that opens, open the diff tool program folder.
4. Select the diff tool .exe file on your computer and press Open. The name of the .exe file is now shown in the Diff tool field (1).
5. Set the Argument (3) field value. The value can vary between diff tools, for example: %1 %2 or diff %1 %2
6. Press Apply (4).

12.3. To compare two workspaces

To start a comparison of two workspaces, go to the top menu Workspace, and select one of the alternatives below to get the diff tool (11) to show the differences.

12.3.1. Compare with saved file

Select Workspace/Compare with saved file to make a comparison of the saved version and the unsaved version of the current workspace.

12.3.2. Compare with other workspace

Select Workspace/Compare with other workspace to make a comparison of the current workspace and another, separate workspace file.

In the window that opens, go to the workspace file you want to compare with, select it and press Open.

12.3.3. Advanced compare with other workspace

Select Workspace/Advanced compare with other workspace to make a comparison of the current workspace and another workspace file where the setup and configurations can be selected for the comparison.

Press Browse (5) to select the other workspace file you want to compare with.

Check the checkboxes (6) to specify what you like to include in the comparison.

The checkbox Include all configurations (7) is checked by default, but you can uncheck it to select specific configurations (8) to include in the comparison for each of the workspaces.

The Close dialog after diff (9) is checked by default, and it closes the window after the comparison is uploaded into the diff tool. Uncheck the checkbox if you want to keep the window open.

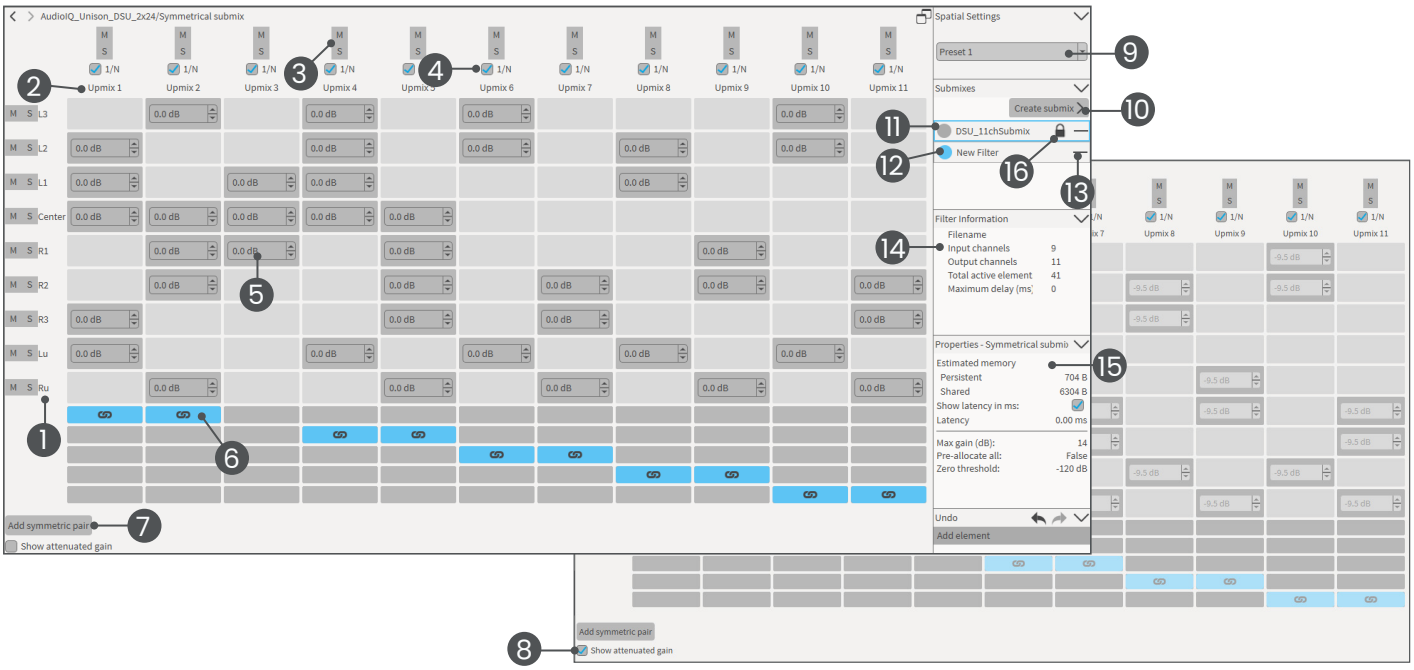
Press Diff (10) to upload the comparison into the Diff tool.

12.4. To compare two configurations

To start a comparison of two configurations in the same workspace, go to the top menu Workspace/ Compare configurations in this workspace.

This opens a specialised version of the Advanced compare dialog, where you can select what to include in the comparison and which configurations to compare.

Press Diff (10) to send the comparison to the Diff tool.



1. Input channel
2. Output channel
3. M = Mute, S = Solo
4. 1/N input distribution
5. Filter cell
6. Symmetric pair
7. Add symmetric pair button
8. Show attenuated gain button
9 Spatial settings
10. Create submix button
11. Filter shown in panels
12. Filter sent to device
13. Remove filter
14. Filter information
15. Properties
16. Locked for editing

13. Symmetrical submixer

13.1. Overview

The **Symmetrical submixer** processor is a type of Matrix mixer, but without delays. Each input channel can be mapped to an output channel and adjusted for gain. The output channels can be paired and set as a symmetric pair. This is also where you can mix Dimensions.

13.2. To create new/import a filter

Press the **Create submix** (10) button. You can choose to:

A. **Import filter**. This option imports a filter for inspection and editing. You can import multiple filters into the view to inspect them one at a time.

B. **Create new**. Create a blank, editable filter and add filter cells, values and symmetrical pairs as described below.

Press the file name to show the filter data in the processor panel (1-8) and properties panel (14-15). The filter that is shown in the panels has a blue border.

To send a specific filter to the device, press the dot (12) before the filter file name. The dot becomes blue for the filter that is sent to the device.

13.3. Spatial settings

Dimensions sets the spatial settings of the symmetrical submixer. A spatial setting must be selected in the **Spatial settings** (9) field. If you have more than one spatial settings, you can select from the drop-down list a specific setting for the symmetrical submixer.

13.4. To edit a filter

You can add and remove filter cells, adjust the values for them, and pair the output so that the two paired output channels become symmetric.

13.4.1. Mute/Solo

Press **M** (3) on the input/output channel(s) you want to mute. You can mute multiple channels.

Press **S** (3) on the input/output channel(s) you want to listen to on their own. This mutes all other channels. You can select multiple channels as Solo channels.

13.4.2. 1/N

If there are two or more filter cells for an output, you can check the checkbox **1/N** (4) to merge the filter cell values for the output channel to prevent clipping.

13.4.3. Filter cell

Click on an unselected, dark grey filter cell (5) in the processor panel to select it. A selected filter cell has a blue border. The cell value is editable both when the cell is selected or unselected.

To move a filter cell, drag it to an unused filter cell and drop it in the cell.

To add a new filter cell, right-click on an unused, light grey filter cell and select **Add element**.

To copy an existing filter cell, right-click it and select **Copy**, then right-click an unused filter cell and select **Paste**. You can also press CTRL, click the filter cell you want to copy, and drag it to an unused filter cell.

13.4.4. Add symmetric pair

Press the **Add symmetric pair** (8) button to add a row for pairing. You can pair 2 output channels per row. An output channel can be paired only once.

In the pairing row, hover over the pairing cell with the mouse pointer. If the pairing cell becomes dark grey, it can be paired.

Click on the first pairing cell for the output channel you want to pair. The cell now has the blue pairing icon (6).

Hover over the second output channel pairing cell to make sure it can be paired. If it becomes dark grey, click on it to pair the second output channel. Both channels now have the pairing icon (6) in the same pairing row.

If you want to remove the pairing, click on the pairing icon to remove the pairing from an output channel. You can now add the pairing to another output channel in the same row.

When two output channels are paired, Flow automatically makes a symmetric setup of filter cells. For each paired output channel, there is a symmetrical setup of filter cells in the channels.

Any changes that are made to one of the paired output channel filter cells are automatically adjusted in the corresponding output channel filter cell in the paired output channel.

Example:

When output channel 1 and output channel 2 are paired:

If you add filter cells to the input channels Left 2 and Center for output channel 1, Flow automatically adds filter cells to input channels Right 2 and Center for output channel 2.

When the gain of Right 2 is adjusted, the gain of Left 2 is automatically adjusted to the same value.

NOTE: If the setup includes uncorrelated Dimensions output channels, these are not included in the pairing for the output and are not part of the automatic symmetrical adjustments.

13.4.5. Show attenuated gain

Check the checkbox **Show attenuated gain** (8) to display the real gain for all filter cells. In this mode, the values of the filter cells and the pairings are not editable.

13.4.6. Lock for editing

Filters can be locked for editing. When you are satisfied with a filter, right-click it and select Lock for editing to prevent unintentional changes. A locked filter shows a padlock icon (16); an unlocked filter has no icon. To make the filter editable again, right-click it and select Unlock.

A filter that is imported from a file is locked by default, but it can be unlocked.

A filter that is imported with Dirac IO is always locked and cannot be unlocked, because it is controlled by the client. If you want to edit the filter, duplicate it to create an editable copy.

13.5. To duplicate a filter

If you want to make an exact copy of a filter, right-click on the filter name and select **Duplicate**. The copy of the filter is shown in the parameters panel, with a number after the file name.

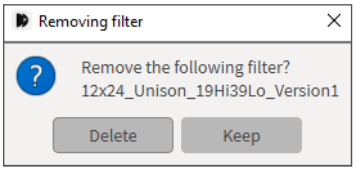
13.6. To export a filter

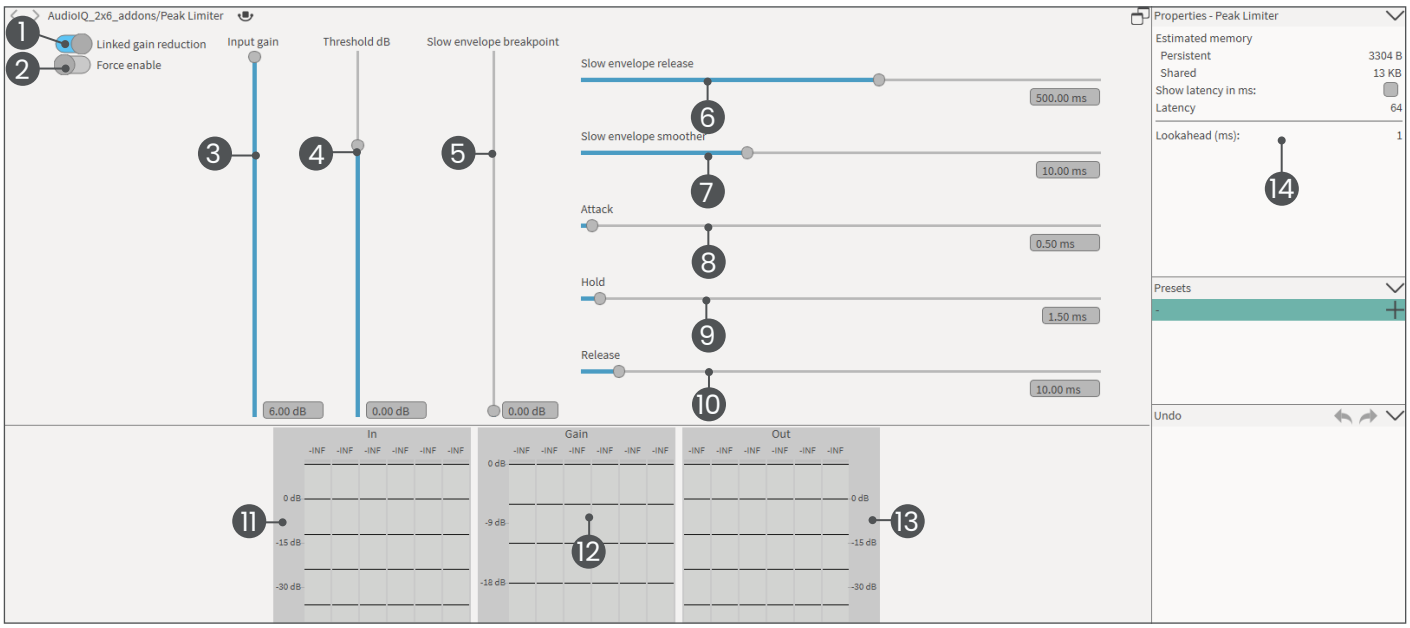
Right-click the filter you want to export (11 or 12). Select the applicable file format for this filter, and save it to your chosen location.

NOTE: Filter export requires an applicable licence.

13.7. To remove a filter

Press the minus symbol (13) on the filter you want to remove from the panels. In the popup window, press **Delete** if you want to remove it, or press **Keep** to cancel.





1. Linked gain reduction
2. Force enable
3. Input gain
4. Threshold dB
5. Slow envelope break point
6. Slow envelope release
7. Slow envelope smoother
8. Attack
9 Hold
10. Release
11. Input meter
12. Gain meter
13. Output meter
14. Properties

14. Peak limiter

14.1. Overview

The **Peak limiter** processor limits the input signal by setting a maximum threshold value that prevents clipping.

14.2. Setup

Set the **Linked gain reduction** (1) to On to link all channels. You can use it for corresponding channels.

If you want to make sure that the Peak limiter is applied and cannot be bypassed, set **Force enable** (2) to On.

The properties (14) show the estimated memory usage and Lookahead value.

14.3. Settings

The **Input gain** (3) sets the signal gain level for the audio playback input.

The **Threshold dB** (4) sets the signal level for when the gain reduction starts. A lower value gives a harder reduction.

The **Slow envelope breakpoint** (5) sets the gain level for when the slow envelope process is started.

The **Slow envelope release** (6) sets the time for when the slow envelope is stopped after the gain reduction is below the breakpoint.

The **Slow envelope smoother** (7) sets a time constant for smoothing the envelope.

NOTE: The slow envelope parameters sets values for the calculated reduction of the gain.

The **Attack** (8) slider sets the time for when the full reduction of the gain should be reached. The attack value must be less than 50% of the Lookahead value for best effect.

The **Hold** (9) slider sets the time for how long the audio should be sustained at the same level before it is released.

The **Release** (10) slider set the time for how long it should take until the gain has returned to 0 dB.

NOTE: Overshoots can still occur if the relation between Attack, Hold, and Lookahead times are not balanced. You can adjust Attack and Hold times as needed to prevent overshoots. The Lookahead time is a fixed value that is set for the signal chain.

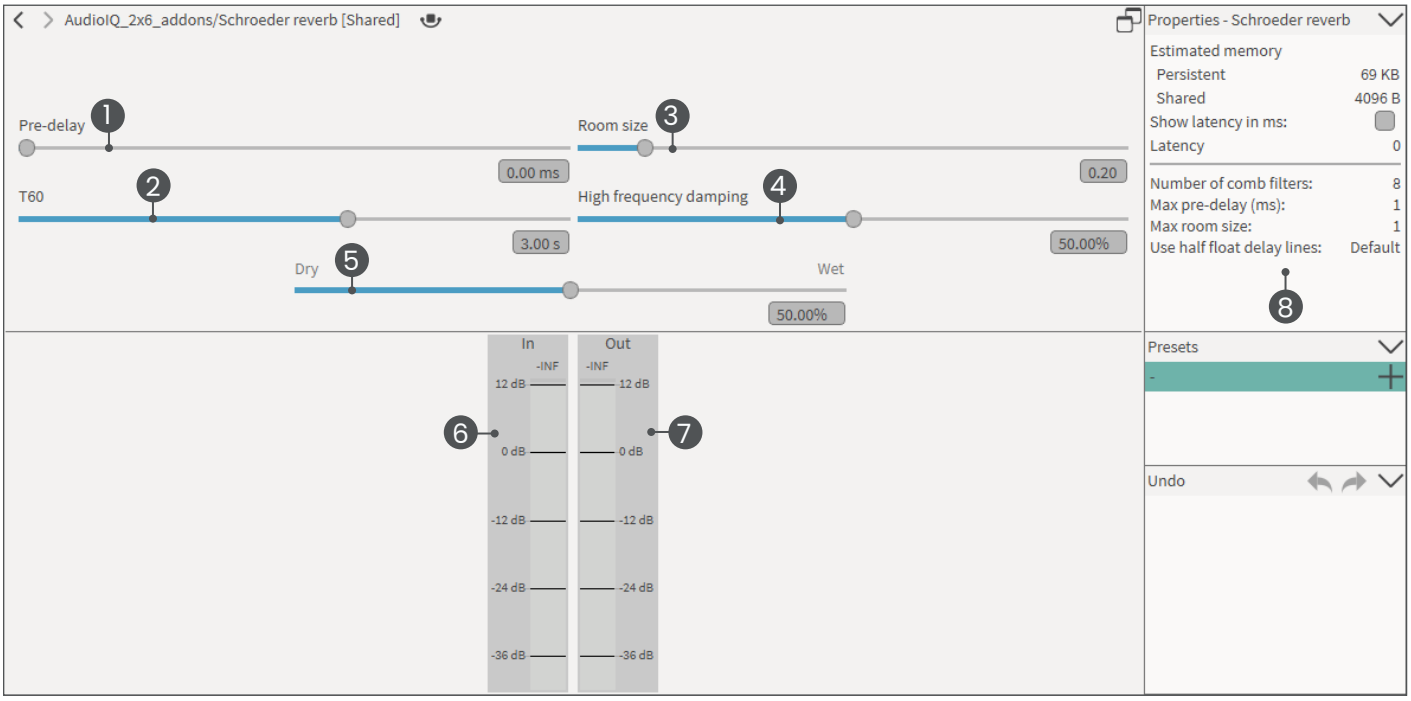
14.4. Level meters

The level meters at the bottom of the processor panel shows:

The Input meter (11) shows the input levels from the playback.

The Gain meter (12) shows the applied gain levels of the peak limiter processing.

The Output meter (13) shows the output levels after processing.



1. Pre-delay
2. T60
3. Room size
4. High-frequency damping
5. Dry/Wet
6. Input meter
7. Output meter
8. Properties

15. Reverb

15.1. Overview

The **Reverb** processor is where the reverb response is calculated according to the Schroeder reverb algorithm.

15.2. Settings

The Pre-delay (1) slider sets the time in milliseconds for the length of the pause before the onset of the reverb signal.

The T60 (2) slider sets the time in which the reverb signal should decrease by -60 dB.

The Room size (3) value corresponds to an estimated room volume for the calculation of the reverb. Setting a higher the value results in a larger room impression of the reverb.

The High frequency damping (4) slider sets the percentage of how much the high frequencies should be dampened.

The Dry/Wet (5) slider sets the balance between the reverb signal and a passthrough of the input signal.

15.3. Level meters

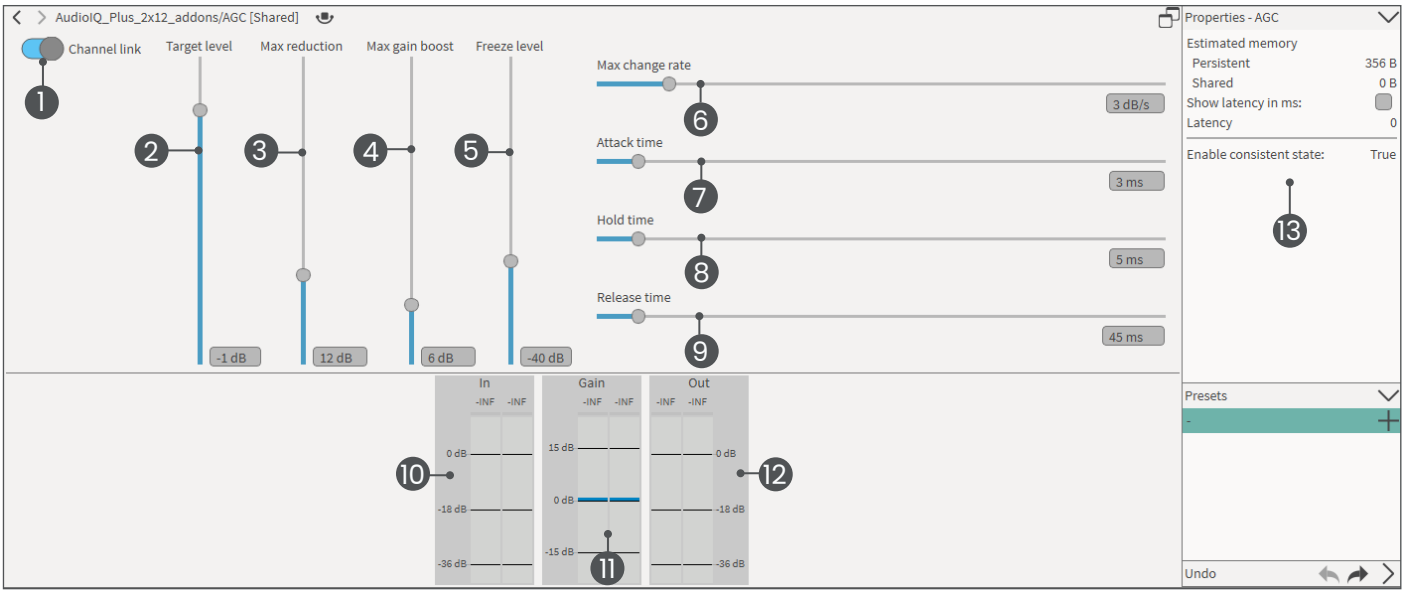
The level meters at the bottom of the processor panel shows meters for:

The Input meter (6) shows the input levels from the playback.

The Output meter (7) shows the output levels after processing.

15.4. Properties

The properties (8) show the estimated memory usage, latency, the number of comb filters, maximum pre-delay, maximum room size and the usage of half float delay lines.



1. Channel link
2. Target level
3. Max reduction
4. Max gain boost
5. Freeze level
6. Max change rate
7. Attack time
8. Hold time
9 Release time
10. Input meter
11. Gain meter
12. Output meter
13. Properties

16. AGC

16.1. Overview

The **AGC (Automatic Gain Control)** processor is where the input signal is analysed and calculated against the target level. The AGC applies slow changes that makes the user perceive the audio at the same gain level.

16.2. Setup

Set the **Channel link** (10) to On to link corresponding input channels (for example Left and Right) to apply the same gain to all channels.

16.3. Settings

The **Target level** (2) slider sets the intended level for the output signal.

The **Max reduction** (3) slider sets maximum permitted gain reduction for the audio.

The **Max gain boost** (4) slider sets the maximum permitted gain increase when there is a weak input signal.

The **Freeze level** (5) slider sets the level for when the AGC processor should stop processing the input signal. Signal levels below the set Freeze level are not processed.

The **Max change rate** (6) slider sets the maximum speed at which changes should be applied for the output.

The **Attack time** (7) controls the attack of the envelope-following of the input signal.

The **Hold time** (8) controls the hold of the envelope-following of the input signal.

The **Release time** (9) controls the release of the envelope-following of the input signal.

16.4. Level meters

The level meters at the bottom of the processor panel shows meters for:

The Input meter (10) shows the input levels.

The Gain meter (11) shows the currently applied gain.

The Output meter (12) shows the output levels after processing.



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